

Documentation Tool

SW11-609

**Implementation
Startup and Configuration**

***Documentation Tool
(Doc Tool)***

**SW11-609
Release 680**

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Table of Contents

1.	INTRODUCTION.....	11
1.1	About This Document	11
	Basics	11
	Document background	11
1.2	Documentation Tool Concept	12
	Definition	12
	Purpose.....	12
	Conceptual diagram	12
	Typical users	13
	Displaying data.....	14
1.3	Before Using the Documentation Tool	15
	Two required procedures	15
	Procedure 1.....	15
	Procedure 2.....	16
	Create Documentation Control and Temporary File Directories.....	16
	Modify Default Volume Pathname display	17
	Create the Query Control File	17
	Documentation Tool display	18
1.4	Comparing Documentation Tool to Find Names.....	19
	Functional differences	19
2.	USING DOCUMENTATION TOOL.....	21
2.1	Overview	21
	Description	21
	This section	21
2.2	Application Examples	22
	Two examples	22
	Example 1	22
	Opening the file	22
	Opened CL source file	22
	Defining a field	23
	Filtering data	23
	Results of the Filter command.....	24
	Outputting data.....	24
	Example 2	24
	Opening the file	24
	Opened Find Names file.....	25
	Defining a field	25
	Sorting data.....	25
	Results of Unit Field Sort, example	26
	Outputting data.....	26
2.3	Accessing the Documentation Tool	27
	Keylock access	27
	Procedure.....	27
	Documentation Tool display	28
2.4	Fields	29
	Definition	29
	Field name example	29
	Creating fields	30
	Field types.....	30
	Date/time formats.....	30

Table of Contents

Changing field types	31
Field name size	31
2.5 Queries	32
Definition	32
Query examples	32
Query operations	32
Query commands	33
Query setup example	33
Query results	33
Query results example	34
Query errors	34
2.6 Command Summary	35
Overview	35
2.7 Function Key Summary	36
Function keys	36
2.8 Documentation Tool Procedures	37
Overview	37
2.9 Opening Files	38
Definition	38
Procedure	38
Pathname entry box	39
2.10 Closing Files	40
Definition	40
Procedure	40
2.11 Finding Data	41
Definition	41
Procedure	41
Pattern and Field entry boxes	41
2.12 Filtering Data	42
Definition	42
Filter state	42
Procedure	42
Filter State and Condition entry box	43
2.13 Sorting Data	44
Definition	44
Sort rules	44
Sort example	44
Procedure	45
Sort State and Field Name entry box	46
Sorting Journal Queries	46
2.14 Outputting Files	48
Definition	48
Keylock level	48
Save options	48
Procedure	50
Output command pathname and save options	50
2.15 Defining, Changing, and Deleting Fields	51
Procedure	51
Field name and Type menu	52
2.16 Building, Saving, Executing and Recalling a Query	53
Finding query- building information	53
Saving query data	53
Saving query for Recall	53
Saving setup data	53

Saving results data.....	54
Recalling a query	54
Before beginning	54
Modify Default Volume Path Name display	55
Procedure.....	55
Query display	57
Query result.....	57
Query Journal Events.....	58
Overview	58
Comparing Query Journal to Event History	58
Query Journal setup display.....	59
Retrieval Filter for Journal Queries.....	59
Query Journal results	62
Query Find Names	62
Overview	62
Query Build Find Names Form Display	63
Query Build Form for Point Interconnection Details	64
Procedure.....	64
Results of a point interconnections search using an FNM query descriptor	66
Examples of setting up pre-built FNM queries.....	67
Event Types	69
2.17 Rules for Entering Query Setup Data	72
List of rules.....	72
Parameters occurring on every point	74
Queries with Selection List.....	74
File Format for Selection List.....	76
2.18 Modifying and Executing a Prebuilt Query	78
Procedure.....	78
Prebuilt Query descriptors menu.....	79
2.19 Deleting a Prebuilt Query	80
Procedure.....	80
2.20 Conditions for Query and Filter Commands	81
Definition	81
Release 610	81
Condition format.....	82
Condition operator precedence	83
Condition with Filter command	84
Condition with Filter command and fieldname on right side of operator	84
Condition using [NOT] operator with Filter command.....	84
Condition with Query command	85
Condition with Query command and parameter name on right side of operator	85
Condition rules	85
Conditions for Query Journal.....	86
2.21 Patterns for Query, Find, and Filter Commands.....	87
Definition	87
Pattern special characters.....	87
2.22 Help Function	88
Overview	88
Accessing the Help function	88
Leaving the Help function.....	88
Example	88
2.23 Responding to Errors	89
How errors appear	89
Error box overlay	89
Error box targets	89
F9 error message	90
Display with F9 error message.....	90

Table of Contents

2.24	Out of Memory/Storage Space Errors.....	91
	Ways to exceed memory or storage space	91
	Exceeding Universal Station memory	91
	Universal Station out of memory error box	91
	Procedure.....	92
	Cancel Error box	92
	Exceeding storage space capacity.....	92
	Removable media out of memory error box	93
	Deleting files.....	93
	Procedure.....	93
2.25	Setting the Directory Pathname	94
	Why set the pathname?	94
	Procedure.....	94
2.26	Error Messages	95
	Overview	95
	Messages.....	95

Tables

Table 1 Procedure 1 - Create Directories and Set Pathnames	16
Table 2 Procedures 2 – Creating a Query Control File.....	17
Table 3 Documentation Tool vs. Find Names.....	19
Table 4 Accessing the Documentation Tool	27
Table 5 Date/Time Formats	30
Table 6 Commands.....	35
Table 7 Function Keys	36
Table 8 Open Commands.....	38
Table 9 Close Commands.....	40
Table 10 Find Command.....	41
Table 11 Filter Command.....	42
Table 12 Before Sort.....	44
Table 13 After Sort.....	45
Table 14 Sort Command	45
Table 15 JRNL SORT Command	46
Table 16 Save Options.....	48
Table 17 Output Command.....	50
Table 18 Defining Fields	51
Table 19 Changing Fields	51
Table 20 Deleting Fields	52
Table 21 Removable Media Step.....	54
Table 22 Build, Save, and Execute a Query	55
Table 23 Retrieval Filter Command	60
Table 24 Search Rules for Entering Query Find Name Data.....	64
Table 25 Defining Point Interconnection Search Parameters, Scenario 1	67
Table 26 Defining Interconnection Search Parameters, Scenario 2.....	68
Table 27 Event Types	69
Table 28 Rules for Entering Query Setup Data	72
Table 29 Parameters Occurring on Every Point	74
Table 30 Execute a Built and Saved Query.....	78
Table 31 Deleting a Built and Saved Query.....	80
Table 32 Condition Operators.....	83
Table 33 Condition Rules.....	85
Table 34 Values	86
Table 35 Special Characters to Use in Patterns.....	87
Table 36 Help File Displays	88
Table 37 Error Message Box Targets.....	89
Table 38 Running Out of Universal Station Memory	92
Table 39 Delete File	93
Table 40 Setting the Directory Pathname	94

Figures

Figure 1 Conceptual diagram.....	12
Figure 2 Modify Default Volume Pathname Display	17
Figure 3 Documentation Tool Display	18
Figure 4 Opened CL Source File	22
Figure 5 Define a Field to Identify CL Statement Types	23
Figure 6 Results of the Filter Command	24
Figure 7 Opened Find Names File.....	25
Figure 8 Results of Unit Field Sort Example	26
Figure 9 Documentation Tool Display	28
Figure 10 File With Field Names Example.....	29
Figure 11 Query Setup Display Example.....	33
Figure 12 Query Setup Display Example.....	34
Figure 13 Pathname Entry Box Display Example	39
Figure 14 Pattern and Field Entry Boxes Display Example	41
Figure 15 Filter State and Condition Entry Box Display Example.....	43
Figure 16 Sort State and Field Name Entry Box Display Example.....	46
Figure 17 Journal Sort for Journal Query Setup	47
Figure 18 Journal Query with Output in CSV format Example	49
Figure 19 Output Command Pathname and Save Options	50
Figure 20 Field Name and Type Menu Display Example	52
Figure 21 Modify Default Volume Path Name Display Example.....	55
Figure 22 Query Display Example	57
Figure 23 Query Result Example.....	57
Figure 24 Query Journal Setup Display Example	59
Figure 25 Retrieval Filter for Journal Query Example.....	61
Figure 26 Query Journal Results Example	62
Figure 27 Query Build Find Names Form	63
Figure 28 Example of Interconnection Result.....	67
Figure 29 Prebuilt Query Descriptors Menu Display Example.....	79
Figure 30 Conditions Format	82
Figure 31 Help Display	88
Figure 32 Error Box Overlay	89
Figure 33 Display with F9 Error Message.....	90
Figure 34 Universal Station Out Of Memory Error Box.....	91
Figure 35 Cancel Error Box.....	92
Figure 36 Removable Media Out Of Memory Error Box.....	93
Figure 37 Volume Specified Not Found Error Message	94

1. Introduction

1.1 About This Document

Basics

Purpose	Describes the functionality of the Documentation Tool.
Intended audience	Operators, maintenance technicians, or engineers who need to retrieve and customize statistical and conditions data from LCN nodes and process-connected devices including AMs, CMs, the UCN, and Data Hiways.
For product release	TPS Release 680.

Document background

This Introduction

- Defines the Documentation Tool Concept and lists functionalities in section 1.2
- Comparing Documentation Tool to Find Names in section 1.4
- Describes procedures to do Before Using the Documentation Tool in section 1.3

The rest of this document will

- Define the many commands available with the Documentation Tool in section 2.6
- Describes the Documentation Tool Procedures required to use these commands in section 2.8
- Gives examples that show how the Documentation Tool can benefit you in your job in section 2.2

1.2 Documentation Tool Concept

Definition

The Documentation Tool is a TDC 3000^X Engineering Personality function that allows you to retrieve data from several sources including.

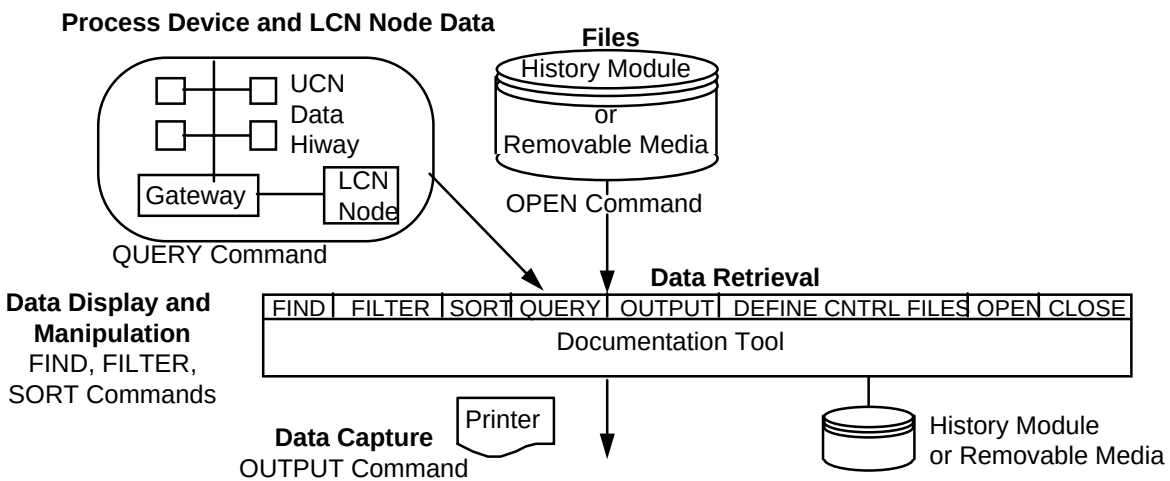
- Process-connected devices
- LCN nodes
- Files from the History Module and removeable media

Purpose

Once retrieved, data can be displayed or manipulated by using various commands within the Documentation Tool, creating more customized information. Results can then be stored to a file or printed. Queries depict current system information and assist you in documenting the system database.

Conceptual diagram

This diagram illustrates how the Documentation Tool retrieves data for queries and from files on storage media. The data can then be manipulated by using the Find, Filter, or Sort commands with the results being output to a file or a printer.



52937

Figure 1 Conceptual diagram

Typical users

Operators, maintenance technicians, or engineers can benefit from Documentation Tool results. The following examples depict how it assists these users.

Operators

Operators can obtain current process conditions on demand. For example, when an operator first starts a shift, a query can be executed listing all LCN points currently in emergency alarm, manual mode, or inactive state. After the query results are displayed, they can be sorted so that points are listed in alphabetical order by unit. Sorted data can then be printed.

Technicians

- Example 1: A query can be created to list current tuning constants for all PID controllers, with the results sorted by hiway box or UCN node. The sorted listing shows tuning constants for each device that can be used to note constants changed by operations personnel or to document tuning after startups.
- Example 2: The configuration of logic points can be traced to find the cause of a tripped interlock by executing a query listing all logic points in a particular device and their corresponding input connections, interlocks, and permissives.
- Example 3: If a site is using the UCN Advanced Process Manager, a query can be executed to obtain statistics such as the number of transitions and total run time for motors being controlled by Device Control or Digital Composite points.

Engineers

Database resources can be documented more easily by performing queries on all hiway boxes or UCN nodes to show which boxes or nodes are presently being used.

Example: One query can retrieve all numerics listed by box or node number, point name, and internal variable number. A second query can retrieve all Regulatory Control points and their corresponding algorithms, input sources, output sources, and units.

The data can be sorted by box or node number to produce more relevant information and then saved to a file. The contents of this file can be edited in the Text Editor to include descriptive comments.

Displaying data

Data is presented in single-or multiple-page displays. Displays show the content of files you opened or the results of queries you executed while in the Documentation Tool. Displays may also appear as boxes of information the Documentation Tool creates automatically, such as an error message box. These boxes overlay the current display.

Each display remains available for recall until it is cancelled or closed. Every new display overlays the previous one, creating a stack of displays. You can page through displays in the stack using the <DISP FWD> and <DISP BACK> keys.

If you try to bring more displays into the stack than Universal Station memory can handle, you are asked to cancel one or more displays.

1.3 Before Using the Documentation Tool

Two required procedures

Before you use the Documentation Tool for the first time, two procedures must be performed:

- Procedure 1- create required directories and set pathnames for them
- Procedure 2 - create a query control file

Procedure 1

Procedure 1 will Create Documentation Control and Temporary File directories and specify their pathnames to the Universal Station. Because this procedure involves using the Command Processor, the keyswitch must be in the ENG position. If your site has users that store these prebuilt queries on different media, Procedure 1 may be required each time the Documentation Tool is accessed from a particular Universal Station.

Purpose of the Directions

The Documentation Control and Temporary File directories created in Procedure 1 serve the following purposes:

1. Documentation Control directory
 - This directory contains a file with the predefined name of SYSQRY.DC. It holds the data generated from query saves.
2. Temporary File directory
 - Temporary files hold data currently being displayed and exist only while a function is in effect. Ordinarily, the Temporary File directory is empty unless an engineering function that uses temporary files is in progress, such as the Command Processor List (LS) command, the Picture Editor, and the Documentation Tool Sort and Query commands.
 - Temporary files are normally deleted after exiting the Documentation Tool, but on occasion, abnormal errors leave them in the directory. You can delete these files if this happens.

Procedure 2

Procedure 2 of the Documentation Tool installation will Create the Query Control File, a control file named SYSQRY.DC. Prebuilt queries are stored in this file. If it is not present, errors result when using the Query command. The Define Cntrl Files command in the Documentation Tool initially creates the SYSQRY.DC file.

Create Documentation Control and Temporary File Directories

The first procedure to perform before using the Documentation Tool is as follows.

Table 1 Procedure 1 - Create Directories and Set Pathnames

Step	Action	Result
1	Put the keyswitch in the ENG position.	The access level is now ENG.
2	Call up the Command Processor display to create the Documentation Control and Temporary File directories. Example: CD NET>VOL &DOC CD NET>VOL TFIL	The directory and pathnames are specified later in the Modify Default Volume Path Name display. NOTE: The default directory names in the pathname catalog are &DOC and TFIL. You may define other names if you wish. The directories can be created on either the HM or removable media.
3	Using different pathnames other than &DOC and TFIL that are defined in pathname catalog? • If yes, perform Steps 4 through 6. • If no, go to Procedure 2.	You must make additional entries. You have completed Procedure 1.
4	Perform one of the following procedures to call up the Modify Default Volume Path Name display. • If Engineering Select Main Menu is on screen, select • If Command Processor display is on screen, Type in SP and press <ENTER>.	The Modify Default Volume Pathname display appears, as shown at the end of this procedure. The Documentation Tool uses the specified pathnames to access the Documentation Control and Temporary File directory.
5	Type in the desired pathnames. Use the directory names and media created in Step 2.	Pathnames for Documentation Control and Temporary File directories are defined.
6	Perform one of the following procedures to return to prior display. • If prior display was Engineering Main Menu, select [MAIN MENU]. • If prior display was Command Processor, hold <CTL> and Press <MENU>.	Prior display appears.

Modify Default Volume Pathname display

DD:MMM:YY HH:MM:SS 1

MODIFY DEFAULT VOLUME PATH NAMES				
Edit all desired default paths and ENTER				
HG GDF NET>&HGG>	NETWORK CONFIG NET>&ASY>	CL OVERLAY NET>&OP2>	DEB OVERLAY NET>&OP1>	SDT OVERLAY NET>&OP4>
HM/AM/CM GDF NET>&AMG>	CL SOURCE/OBJ NET>CL>	PICTURE EDITOR NET>&OP2>	LBC OVERLAY NET>&OP1>	FIND NAMES OVLY NET>&OP4>
AREA DB GDF NET>&ARG>	CL PARAM LIST NET>CL>	FFL OVERLAY NET>&OP2>	TRANSLATORS OVL NET>&OP4>	LOAD NODE OVRLY NET>&OP4>
CL CUSTOM GDF NET>&CDSG>	USER DEFLT PATH NET>TEST>	BUTTN CFG OVRLY NET>&OP1>	CONFIG OVRLY NET>&OP1>	GENERIC OVRLAYS NET>&OVG>
NIM GDF NET>&NMG>	KEY FILE VOLUME NET>&KFO>	SMCC OVERLAY NET>&OP2>	TAC SUPPORT OVLY NET>&OP5>	
NIM GDF NET>&NM2>	EXT LOAD MODULE NET>&CUS>	DOC CTL DIR NET>&DOC>	TEMP FILE DIR NET>TFIL>	NCF BACKUP PATH NET>&HGG>
SET DEVICE PATH TO REM. MEDIA	SET DEVICE PATH TO "NET"	MAIN MENU	UTILITIES MENU	

52938

Pathname for Documentation Control directory

Pathname for Temporary File directory

Figure 2 Modify Default Volume Pathname Display

Create the Query Control File

The second procedure to perform before using the Documentation Tool is as follows.

Table 2 Procedures 2 – Creating a Query Control File

Step	Action	Result
1	Perform one of the following procedures to access the Documentation Tool. - If Engineering Main Menu is on screen select. - If Command Processor display is on screen, type in DT and press <ENTER>.	Documentation Tool display appears, as shown at the end of this procedure. Commands are displayed across the top of the screen as targets to be selected. Additional functions available as keys are displayed across the bottom of the screen.
2	Select [DEFINE CNTRL FILE] and press <ENTER>.	The SYSQRY.DC file is created in the Documentation Control directory, then the message "Operation complete" appears. You can now use either the Open or Query commands in the Documentation Tool. For command procedures see 2.9 Opening Files. or 2.16 Building, Saving, and Executing a Query.

Documentation Tool display

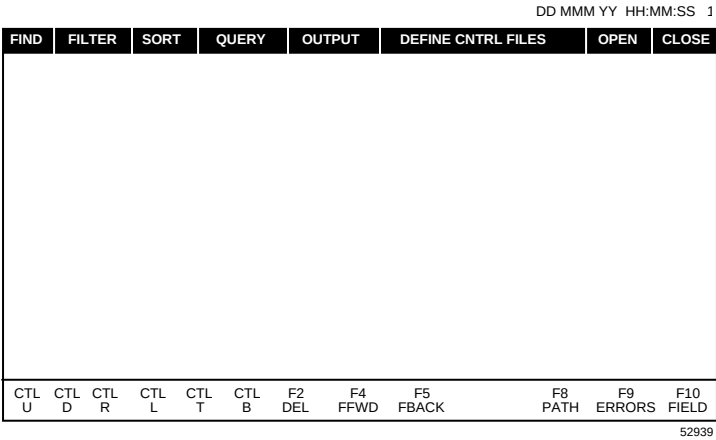


Figure 3 Documentation Tool Display

1.4 Comparing Documentation Tool to Find Names

Functional differences

The following table depicts how the Documentation Tool differs from the Find Names (FN) function of the Command Processor.

Table 3 Documentation Tool vs. Find Names

Function	Documentation Tool	Find Names
Searches the following files for cross references to entities or patterns: History Unit, Area Database, Picture Editor, Button Source, checkpoint, and text files.		√
Searches Display databases (DDB).		√
Lists the file name that contains the character string you are searching for, and prints out the string.		√
Searches the on-line databases of LCN nodes and process-connected devices including: AMs, CMs, the UCN, and Data Hiways	√	
Displays the entire contents of a file once the specified character string is found.	√	
Saves engineering effort and time by arranging information in a useful manner.	√	

2. Using Documentation Tool

2.1 Overview

Description

The Documentation Tool saves engineering time and effort by arranging information in a useful manner. With it you can:

- Search the on-line databases of LCN nodes and process-connected devices including:
AMs, CMs, the UCN, and Data Hiways.
- Display the entire contents of a file once the specified character string is found.
- Customize system database documentation so it can be stored or printed.

This section

- Defines the Commands available in section 2.3
- Describes the Documentation Tool Procedures required to use these commands in section 2.8
- Gives examples that show how the Documentation Tool can benefit you in your job in section 2.2, and
- Provides procedures for Accessing the Documentation Tool in section 2.3

2.2 Application Examples

Two examples

The following examples depict the breadth of Documentation Tool functionality, and also help you gain insight into how Documentation Tool commands are implemented.

Specifically,

- Example 1 is a step-by-step approach of Opening the file, Defining a field, and using the Results of the Filter command to display only certain data.
- Example 2 is also a step-by-step approach of opening a file, defining a field, and using the Sort command (Sorting data) within the file in chronological order.

Example 1

Purpose: List all SEND STATMENTs used in this CL program.

Description: You must create a field in the CL source file for the purpose of searching the data contained within the field.

Opening the file

First, you use the Open command to retrieve the CL source file from storage media. To do this, select [OPEN] from the command line, and enter the pathname of the file, for example, NET>TEST>ABMIX000 . CL. This results in the CL source file shown in the following figure to be displayed.

Opened CL source file

DD MMM YY HH:MM:SS 1

FIND	FILTER	SORT	QUERY	OUTPUT	DEFINE CNTRL FILES	OPEN	CLOSE				
<pre>SEQUENCE ABMIX000 (aPM: POINT REACT000) EXTERNAL FIC21000, TIC21000, FY21000, FY22000, FY23000, LI24000, FVL21000, & FVL22000, DVL23000, AG24000, STATE000, & INGA000, FULMT000, CLNDT000, AGTIM000, SIMLT000 -- ANY INTERNAL VARIABLE ASSIGNMENTS TO BE PLACED HERE -- PHASE SETUP STEP VALVE SET FY21000.COMMAND, FY22000.COMMAND, FY23000.COMMAND=RESET SET FVL21000.MODATTR, FVL22000.MODATTR, DVL23000.MODE = MAN & PROGRAM SET FY21000.COMMAND, FY22000.MODATTR, DVL23000.MODATTR, AG24000.MODATTR= STEP SHUT CLOSED FVL21000, FVL22000, DVL23000 OFF AG24000 -- SET FL(001) = OFF STEP FLOW SET FIC21000.MODATTR=PROGRAM STEP MODMAN SET MODMAN STEP OP SET FIC21000.OP=0</pre>											
CTL	CTL	CTL	CTL	CTL	CTL	F2	F4	F5	F8	F9	F10
U	D	R	L	T	B	DEL	FFWD	FBACK	PATH	ERRORS	FIELD

5294

52940

Figure 4 Opened CL Source File

Defining a field

Second, you must define a field to identify CL statement types so data can be searched in that field. To do this, begin by positioning the cursor where you want the end of the field to be and press <CTL> <F10>. A Field name and Type menu appears. Type in the field name. In the following example, STATMENT is the field name used. Then select the field type. In the example shown, the STRING field-type target is selected. When you press the [ENTER] key, a STATMENT field appears at the top of the CL file and a vertical line showing the width of the new field appears, as the figure illustrates.

DD MMM YY HH:MM:SS 1

FIND	FILTER	SORT	QUERY	OUTPUT	DEFINE CNTRL FILES	OPEN	CLOSE
STATMENT SEQUENCE ABMIX000 (aPM; POINT REACT000) EXTERNAL FIC21000, TIC21000, FY21000, FY22000, FY23000, LI24000, FVL21000, & FVL22000, DVL23000, AG24000, STATE000, & INGA000, FULMT000, CLNDT000, AGTIM000, SIMLT000 -- ANY INTERNAL VARIABLE ASSIGNMENTS TO BE PLACED HERE -- PHASE SETUP STEP VALVE SET FY21000.COMMAND, FY22000.COMMAND, FY23000.COMMAND=RESET SET FVL21000.MODATTR, FVL22000.MODATTR, DVL23000.MODE = MAN & PROGRAM SET FY21000.COMMAND, FY22000.MODATTR, DVL23000.MODATTR, AG24000.MODATTR= STEP SHUT CLOSED FVL21000, FVL22000, DVL23000 OFF AG24000 -- SET FL(001) = OFF STEP FLOW SET FIC21000.MODATTR=PROGRAM STEP MODMAN SET MODMAN STEP OP SET FIC21000, OP=0							
CTL	CTL	CTL	CTL	CTL	CTL	F2	F4
U	D	R	L	T	B	DEL	FFWD
							F5
							FBACK
							F8
							F9
							F10
							PATH ERRORS
							FIELD

53409

Figure 5 Define a Field to Identify CL Statement Types

Filtering data

Third, use the Filter command to eliminate all data in the CL program except the SEND statements from the display. Enter the following condition in the entry port after selecting [FILTER] from the command line, and turning the Filter state [ON]:

STATMENT=SEND

As a result, only the records with SEND in the STATMENT field are displayed as shown in the following figure.

Results of the Filter command

DD MMM YY HH:MM:SS 1

FIND	FILTER	SORT	QUERY	OUTPUT	DEFINE CNTRL FILES	OPEN	CLOSE				
STATEMENT											
SEND (WAIT);"READY THE REACTOR"											
SEND (WAIT);"ENTER NEW TIC21000 SETPOINT (RANGE= 0_100 DEGC)"											
SEND ;"MIXTIME WILL BE",AGTIM000.SP," SECONDS"											
SEND ;"BATCH COMPLETE"											
SEND (WAIT);"TANK A VALVE FAILED: REPAIR AND CONFIRM"											
SEND ;PROGRAM RESTART AT PHASE LOADA"											
SEND (WAIT);"TANK B VALVE FAILED: REPAIR AND CONFIRM"											
SEND ;"BATCH RESUMES AT PHASE LOADB"											
SEND ;"Batch not satisfactory; recycle product."											
SEND ;"EXCESS SOLUTION",EXCESS,"GALLONS"											
CTL U	CTL D	CTL R	CTL L	CTL T	CTL B	F2 DEL	F4 FFWD	F5 FBACK	F8 PATH	F9 ERRORS	F10 FIELD

52941

Figure 6 Results of the Filter Command

Outputting data

Fourth, you can save data results to a file or print it. To do this, select [OUTPUT] from the command line.

Example 2

Purpose: Sort a Find Names (FN) file in chronological order.

Description: From the Documentation Tool, you open a Find Names file that was created using the Find Names command. The file includes data Find Names has found searching the database, checkpoints, and references. To sort the data in this file, you must first create one or more fields.

For more information on the Find Names command see the *Find Names Command* document.

Opening the file

First, you use the Open command to retrieve the Find Names file. To do this, select [OPEN] from the command line, and enter the file pathname, for example, \$F1>TEST>TEMP.XX. The Find Names file shown in the following figure is displayed.

Opened Find Names file

DD MMM YY HH:MM:SS 1

FIND	FILTER	SORT	QUERY	OUTPUT	DEFINE CNTRL FILES	OPEN	CLOSE
FN AM_CP MEDIA	ENTITY AMCP	* CL RAMP*	ENTITY	CL			
NET>F94	NO	RAMP304	RAMPAM				
NET>F93	NO	RAMP303	RAMP				
NET>F93	NO	RAMP301	RAMP				
NET>F93	NO	RAMP302	RAMP				
NET>F93	NO	RAMP305	RAMP				
NET>F93	NO	RAMP306	RAMP				
NET>F93	NO	RAMP307	RAMP				
NET>F93	NO	RAMP310	RAMPAM				
NET>F93	NO	RAMP309	RAMP				
NET>F93	NO	RAMP312	RAMP				
NET>F93	NO	RAMP311	RAMP				
NET>F93	NO	RAMP308	RAMP				
NET>F93	NO	RAMP313	RAMP				
NET>F93	NO	RAMP314	RAMPAM				
NET>F93	NO	RAMP315	RAMPAM				
NET>F93	NO	RAMP316	RAMPAM				
NET>F93	EA	RNDNO332	RAMPAM				
NET>F95	SO	RNDNO333	RAMPAM				
NET>F96	WE	RNDNO331	RAMPAM				
FIND NAMES COMPLETE							
CTL CTL	CTL CTL	CTL CTL	F2	F4 F5	F8 F9	F10	
U D	R L	T B	DEL	FFWD FBACK	PATH ERRORS	FIELD	

52942

Figure 7 Opened Find Names File

Defining a field

Second, you must create a field in the file so that data can be sorted. In this example, the field name being defined is UNIT; and STRING is selected as the data type. Other fields defined in this example include X (because we are not concerned with the data in this column), ENTITY, and CL_BLOCK.

Sorting data

Third, you select the Sort command target and enter UNIT as the field name to sort on. When you press the [ENTER] key, the Sort command reorders the units data in chronological order.

Results of Unit Field Sort, example

DD MMM YY HH:MM:SS 1

FIND	FILTER	SORT	QUERY	OUTPUT	DEFINE CNTRL FILES	OPEN	CLOSE
X			UNIT	ENTITY	CL_BLOCK		
MEDIA			AMCP	ENTITY	CL		
-----			-----	-----	-----		
NET>F94			EA	RNDNO332	RAMPAM		
NET>F93			NO	RAMP301	RAMP		
NET>F93			NO	RAMP302	RAMP		
NET>F93			NO	RAMP303	RAMP		
NET>F93			NO	RAMP304	RAMP		
NET>F93			NO	RAMP305	RAMP		
NET>F93			NO	RAMP306	RAMP		
NET>F93			NO	RAMP307	RAMPAM		
NET>F93			NO	RAMP308	RAMP		
NET>F93			NO	RAMP309	RAMP		
NET>F93			NO	RAMP310	RAMP		
NET>F93			NO	RAMP311	RAMP		
NET>F93			NO	RAMP312	RAMP		
NET>F93			NO	RAMP313	RAMPAM		
NET>F93			NO	RAMP314	RAMPAM		
NET>F93			NO	RAMP315	RAMPAM		
NET>F93			NO	RAMP316	RAMPAM		
NET>F95			SO	RNDNO333	RAMPAM		
NET>F96			WE	RNDNO331	RAMPAM		
FN AM_CP * ENTITY * CL RAMP*							
CTL	CTL	CTL	CTL	F2	F4	F5	F8
U	D	R	L	DEL	FFWD	FBACK	PATH

52943

Figure 8 Results of Unit Field Sort Example

Outputting data

Fourth, you can save data results to a file or print it. To do this, select [OUTPUT] from the command line.

2.3 Accessing the Documentation Tool

Keylock access

All commands in the Documentation Tool are available at the Operator keylock level except the Output to File command, and deleting prebuilt queries. Operators can only output data to a printer.



ATTENTION

You can access the Documentation Tool once you have completed the installation procedures as described in 1.3 Before Using the Documentation Tool.

Once the Documentation Tool has been accessed, you use only the Open or Query commands.

References:

2.9 Opening Files

2.5 Queries

Procedure

The following procedure describes the steps required to access the Documentation Tool.

Table 4 Accessing the Documentation Tool

Step	Action	Result
1	Select [DOCUMENTATION TOOL] target from Engineering Main Menu. or Type DT in the Command Processor and press <ENTER>.	Documentation Tool display appears. A sample Documentation Tool display appears at the end of this procedure.
2	Do you want to open a system file? • If yes, select the [OPEN] target. • If no, go to Step 3.	Entry box appears for file pathname. Reference: 2.9 Opening Files
3	Do you want to execute a query? • If yes, select the [QUERY] target. • If no, procedure is concluded.	Reference: 2.16 Building, Saving, and Executing a Query.

Documentation Tool display

Commands are displayed across the top of the screen as targets to be selected.
Additional functions available as keys are displayed across the bottom of the screen.

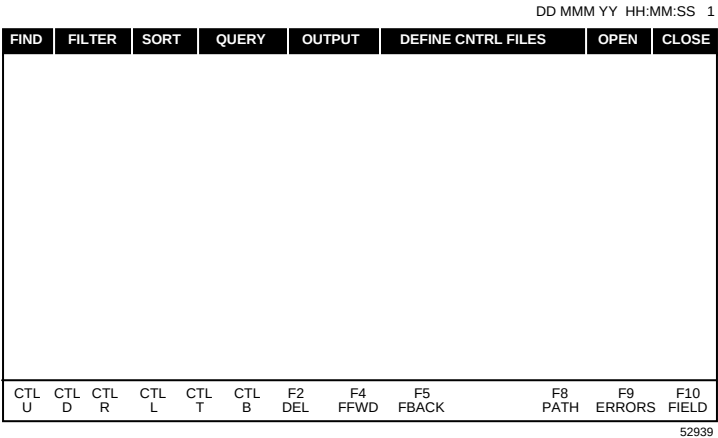


Figure 9 Documentation Tool Display

2.4 Fields

Definition

Fields are columns of data displayed in an open file. You may label these columns with a name and assign them a type to be used in subsequent commands of the Documentation Tool. Once you define a field, the field name is shown at the top of the display, along with vertical lines indicating field boundaries. The following Documentation Tool commands use fields:

- Find
- Filter
- Sort
- Query

Field name example

The following figure depicts a file with field names. In this example, the field names are: ENTITY, PV, SP, MODE, and PTDESC.

FIND	FILTER	SORT	QUERY	OUTPUT	DEFINE CNTRL FILES	OPEN	CLOSE
ENTITY		PV		SP	MODE	PTDESC	
FVL23911		ON		52	AUTO	FLAGFOR PV232911	
FVL24911		OFF		60	AUTO	FLAGFOR PV232911	
DVL23911		ON		32	MAN		
PI23911		OFF		44	CAS	NUMERIC FOR PV232911	
LI24911		50.000000000		68	CAS	STIMULATION FORPUMP911	
FI23911		-----		70	AUTO	FLAGFOR PV232911	
VLV23911		OFF		57	AUTO	DISCHARGE PUMP	
PV23911		ON		25	MAN	LOGIC TOGGLES SOE POINT	
LOGIC FORSOE		62.000000000		75	MAN	SEQUENCE OF EVENTS POINT	
DIS23911		OFF		68	AUTO	FLAG STARTS SOE TOGGING	
FLAG15		OFF		81	AUTO	REACTOR LEVEL INDICATOR	
LI24000		-----					
ASIM000		ON		20	MAN	SOL. A FLOW SIMULATOR	
BSIM000		OFF		30	NAN	SOL. B FLOW SIMULATOR	
DSIM000		OFF		45	MAN	DRAIN FLOW SIMULATOR	

52944

Figure 10 File With Field Names Example

Creating fields

Fields are either created or generated in the following ways:

- The user creates fields by positioning the cursor in the displayed data at the desired end of the field and pressing the <CTL> <F10> keys (Mark Field).
- The Query command results generate fields. The Query command allows entry of entity names/entity list file and parameters in determining what it searches for. When the query execution process is completed, field names and their boundary lines are displayed, along with columns of data.

Field types

Available field types are:

- Strings (ASCII text)
- Numbers (integer, real, exponential formats)
- Date/time

Date/time formats

The date/time formats supported by the Documentation Tool are listed in Table 5.

Table 5 Date/Time Formats

Type	Format(s)	Example												
Duration	DDDD hh:mm:ss NOTE: Only one blank is allowed between days and hours. All three parts of the time must be specified.	21 10:00:33												
Numeric Date	MM/DD or MM/DD/YY MM-DD or MM-DD-YY	01/03/92 01-03-92												
ISO Date	YY/MM/DD YY-MM-DD	92/01/03 92-01-03												
European Date	DD/MMM/YY DD-MMM-YY	13/AUG/92 13-AUG-92												
Time	HH:MM HH:MM:SS HH:MM:SS:NNNN	10:00 10:00:03 10:00:03:250												
Date and Time	Any of the previous date formats with the addition of time. @ - x ~ - x NOTE: @ – specifies time interval without rounding. ~ – specifies time interval to be rounded off to the nearest whole hour. x – specifies time interval for which the output is required.	01/03/92 10:00:03 13/AUG/92 10:00 @ - 8 ~ - 8 <table> <tr> <th><u>Current Time</u></th> <th><u>Symbol Usage</u></th> <th><u>Displayed</u></th> </tr> <tr> <td>11:29:15</td> <td>@ - 8</td> <td>@ - 8</td> </tr> <tr> <td>12:22:04</td> <td>~ - 8</td> <td>~ - 8</td> </tr> <tr> <td>12:35:11</td> <td>~ - 8</td> <td>~ - 8</td> </tr> </table>	<u>Current Time</u>	<u>Symbol Usage</u>	<u>Displayed</u>	11:29:15	@ - 8	@ - 8	12:22:04	~ - 8	~ - 8	12:35:11	~ - 8	~ - 8
<u>Current Time</u>	<u>Symbol Usage</u>	<u>Displayed</u>												
11:29:15	@ - 8	@ - 8												
12:22:04	~ - 8	~ - 8												
12:35:11	~ - 8	~ - 8												



ATTENTION

The @ or ~ symbol can be used only once in a port. Using both symbols in the same port at the same time will result in error messages.

Changing field types

When a query is executed, the field type matches the data type found for that parameter. In some instances a field may contain mixed types of data. You can change the field type by using the <CTL> <F10> keys.

Example: A PV value could be a number or an ON/OFF string. When more than one data type is returned by the query, the type becomes a string. The field “PV” could be changed to a number field type, allowing the Sort command to sort PV by value. Any strings such as ON or OFF will go to the end of the sort.



ATTENTION

Enumerations such as MAN, AUTO, and CAS, and self-defined enumerations such as ON/OFF are considered STRING field types.

Field name size

You can enter up to sixteen characters for a field name.

2.5 Queries

Definition

The Query command retrieves user-created entity data (not system or “\$” entities) from the on-line database. On-line data resides in the following devices:

- Gateways
- Application Module
- Computing Module
- Process-connected devices

Query examples

Examples of the kinds of queries you can create are described as follows:

Example 1

List all points in Unit 1 that are in manual mode, and display their descriptor, PV, and alarm state.

Example 2

List all points in Units 1 through 5 that are in alarm state, and display their description, PV, mode, and alarm state.

Example 3

List all points on process networks 10 and 12 that have tagnames starting with “FIC,” and show their PV, SP, and mode.

Example 4

List all AM regulatory, switch, and custom points in Units 1, 2, and 3 that have a PV greater than 60. For every point listed, show the parameters ALENBST, PTEXECST, and PVEUHI.

Example 5

List all regulatory points on LCN nodes 1-64.

Example 6

List all points from the entity list file in Units 1, 2 and 3 that are in alarm state.

Query operations

Before executing a query, you must first build the query by entering “setup” data. After building the query, you execute it to retrieve the data. The “setup” data that you enter can be saved for future use; this is called a prebuilt query. Setup data includes the following:

- Type of database: hiway, UCN, unit, or node
- Specific entities/entity list file and conditions (optional)
- Parameters to display in the query results (optional)
- Descriptor (required only if saving query “setup” data)
- Entity types

Query commands

The four Query command operations are:

- **Build** — Calls up targets and ports to build, save, and execute a query.
- **Select Pre-built** — Calls up a list of previously built queries from which you can modify or execute a specific prebuilt query.
- **Delete Pre-built** — Calls up a list of previously built queries from which you can delete a specific prebuilt query.
- **Recall Last Query (R680)** — Calls up the last executed query, which you can edit and execute again.



ATTENTION

The recall operation is applicable only to HWY, UCN, UNIT, NODE and JRNL queries. It is not applicable to the queries that are executed in the background. For example, the recall option is not applicable to QPRINT, QFILE, and FNM.

Query setup example

The following figure is an example of the query setup display that searches for all of the entities starting with “FIC” on process networks 10 and 12, and whose PVs and outputs are greater than zero.

FIND	FILTER	SORT	QUERY	OUTPUT	DEFINE CNTRL FILES	OPEN	CLOSE
Build		Hwy					
Select Pre-Built		UCN					
Delete Pre-Built		Unit					
		Node					
		Jrnl					
				Save	Overwrite		
Proc Net List? 10,12							
Device List?							
Entity Names? FIC*							
Conditions: (PV>0.0) AND (OP>0.0)							
Param values to show: PV, SP, MODE							
Descriptor?							
Resource /Entity Types:							
ALL	DIG IN	DIG OUT	DIG COMP	ANL IN	ANL OUT	LOGIC	DEV CTL
REG PV	REG CTL	FLAG	TIMER	NUMERIC	ARRAY	PROC MOD	

52945

Figure 11 Query Setup Display Example

Query results

All parameters listed in the “Param values to show” port of the query setup are displayed in the query results as shown in the following figure.

Parameter names become field names in the query results. Field names are subsequently used by the Find, Filter, and Sort commands. In the following figure, four fields and field names were automatically generated: ENTITY is a string field type, PV and SP are number field types, and MODE is a string field type. The results of the query can be sent to a file or printer by using the Output command.

Query results example

QUERY				
ENTITY	PV	SP	MODE	
FIC101	50.5	50.8	AUTO	
FIC102	55.2	59.1	CAS	
FIC103	100.0	100.0	AUTO	
FIC202	25.4	30.5	CAS	
FIC204	33.4	52.2	MAN	
FIC301	24.4	21.3	MAN	
FIC303	75.6	75.5	AUTO	
FIC401	-----	0.0	MAN	
FIC404	56.5	56.6	CAS	

52946**Figure 12 Query Setup Display Example****Query errors**

The following errors may appear in the data columns of the query results:

?????? Communication on the LCN is disrupted,
or a device has failed (node, hiway, or
box is down).

@@@@@@ Parameter was not configured on the
point, or the entity does not exist. You
can usually correct these errors.

----- Value of parameter has been set to NaN
(Not a Number).

!!! Parameter does not exist on the point.

You are not required to do any error recovery if these errors appear in your data.

2.6 Command Summary

Overview

The Documentation Tool contains several commands that are described in the following table.

Table 6 Commands

Command	Description
[FIND]	Looks for the next occurrence of a specified pattern.
[FILTER]	Displays only data which meets a specified condition.
[SORT]	Reorders the data according to values in the fields.
[QUERY]	Performs a query or alters the prebuilt query file: • [BUILD] Creates and executes queries, which can be saved to a file and printed. • [SELECT PRE-BUILT] Selects a previously built query. • [DELETE PRE-BUILT] Deletes a previously built query. • [RECALL LAST QUERY] Recalls the last executed query (R680).
[OUTPUT]	Sends display data to a file or printer. NOTE: Results displayed from the Query command can also be saved using the Output command.
[DEFINE CNTRL FILES]	Creates the SYSQRY.DC file which holds the data generated from query saves.
[OPEN]	Displays the contents of a file after entry of a pathname. Example: NET>TEST>TEST.EB NOTE: You can use the Command Processor List (LS) command to list available text files, for example: .CL, .EB, .XX, .YY, .ZZ. To go to the Command Processor display, press [ESC]. To return to the Documentation Tool, hold <CTL> and press <MENU>.
[CLOSE]	Removes display currently on the screen.

2.7 Function Key Summary

Function keys

The Documentation Tool contains several function keys that are described in the following table.

Table 7 Function Keys

Key	Description
<PAGE FWD> or <PAGE BACK>	Pages forward and backward through data one full screen at a time.
<DISP FWD> or <DISP BACK>	Pages through the displays in the Documentation Tool. Multiple displays can be invoked, with each overlaying the previous display.
<CTL> <U> or <D>	Scrolls the screen up or down one line at a time.
<CTL> <R> or <L>	Scrolls data right or left across the screen, displaying the next or previous field. If no fields are defined, 10 characters are shifted at a time. Up to 240 characters are displayed per line.
<CTL> <T> or 	Positions display at start (top) or end (bottom) of the data.
<CANCEL>	Removes display currently on the screen. If an unentered command is on the screen, the command is deleted. If a file is on the display, it is closed.
<CTL> <F2> DEL	Deletes a single prebuilt query descriptor line.
<CTL> <F4> FFWD	Repeats a “find” in the forward direction. It searches forward from the current cursor position, using the last pattern entered in the Find command. If a match is found, the cursor is moved to the location of the match.
<CTL> <F5> FBACK	Repeats a “find” in the backward direction. It searches backward from the current cursor position, using the last pattern entered in the Find command. If a match is found, the cursor is moved to the location of the match.
<CTL> <F8> PATH	Displays the pathname of the data where the cursor is currently located. In a display resulting from a query or sort, the system-generated pathname is shown. Example: NET>TEST>26661400.___
<CTL> <F9> ERRORS	Displays errors from the last operation on a separate display. • If 1 or 2 errors are present, then errors are shown in the prompt area. • If more than 2 errors are present, then a status message directs user to use <CTL> <F9> keys. Error messages are present and remain as a separate display on the display stack until cancelled. Multiple <F9> displays may be on display stack.
<CTL> <F10> FIELD	Reads the current cursor position and calls up a menu to define a field name, type, and boundary.

2.8 Documentation Tool Procedures

Overview

The procedures that are discussed in the following subsections include:

- 2.9 Opening Files — Displays a file as defined by the pathname entered in the entry port.
- 2.10 Closing Files — Removes the display from the display stack.
- 2.11 Finding Data — Locates the next occurrence of a pattern or specified field within the current display data.
- 2.12 Filtering Data — Displays a subset of the data currently on the screen.
- 2.13 Sorting Data — Arranges records with a file in ascending order.
- 2.14 Outputting Files — Prints or saves the results of previous Documentation Tool operations such as queries and sorts.
- 2.15 Defining, Changing, and Deleting Fields — Labels columns of data with a name and a type to be used in subsequent operations.
- 2.16 Building, Saving, and Executing a Query — Retrieves user-created entity data from the on-line database.
- 2.19 Modifying and Executing a Prebuilt Query — Modifies a previously built query.
- 2.20 Deleting a Prebuilt Query — Deletes a previously built query.

2.9 Opening Files

Definition

The Open command displays a file as defined by the pathname entered in the entry port. More than one file may be open at the same time, creating a display stack. Once a file is open, you can use either the Find, Filter, Sort, or Define Field commands to manipulate data within the file. Files with lines 80 characters in length can be displayed. You can use the <CTL> <R> or <CTL> <L> keys to scroll right or left to view the data in the file.

Procedure

The following procedure describes the steps required to use the Open command.

Table 8 Open Commands

Step	Action	Result
1	Select the [OPEN] target from the command line.	The pathname entry box appears. A sample Pathname entry box appears at the end of this procedure.
2	Type in the desired pathname and press [ENTER]. Example: NET>TEST>TEST.EB NOTE: If only the file name is entered, the device and directory is obtained from the user default pathname (entered by using the Default Volume Pathname display or the Command Processor SP command).	File is displayed, or an error message appears. If an error message is shown, see 2.23 Responding to Errors.
3	Select another command (either Find, Filter, Sort, or Define Fields) to manipulate data within the opened file, or select the [OPEN] target again to display another file.	The entry box for selected command is displayed. For command procedures see: 2.11 Finding Data 2.12 Filtering Data or 2.13 Sorting Data 2.15 Defining, Changing, and Deleting Fields.

Pathname entry box

FIND	FILTER	SORT	QUERY	OUTPUT	DEFINE CNTL FILES	OPEN	CLOSE
Pathname							

52947

Figure 13 Pathname Entry Box Display Example

2.10 Closing Files

Definition

Closing a file removes it from the display stack.

Procedure

The following procedure describes the steps required to use the Close command.

Table 9 Close Commands

Step	Action	Result
1	Select the [CLOSE] target from the command line. or Press the <CANCEL> key. NOTE: In Free Format Logs (when built) and the Picture Editor, the <CANCEL> key undoes the previous command. In the Documentation Tool, the <CANCEL> deletes the display currently on the screen.	The file is removed from the display stack.

2.11 Finding Data

Definition

The Find command locates the next occurrence of a pattern, or specified field within the current display data. Once you press <ENTER> to start the search, the cursor is positioned over the first character at the beginning of the page and starts the search from, but not including, the first character.

Procedure

The following procedure describes the steps required to use the Find command.

Table 10 Find Command

Step	Action	Result
1	With a file displayed on the screen, select the [FIND] target from the command line.	Both Pattern and Field entry boxes overlay the display. A sample of Pattern and Field entry boxes appear at the end of this procedure.
2	Type in the pattern to be searched. You also have the option of searching on a field. Select a search direction, and press <ENTER>. References: Conditions for Query Journal 2.15 Defining, Changing, and Deleting Fields NOTE: To abort the search, press <CTL> <BREAK>.	“match found” appears and cursor is positioned at location of match, or “no match found” appears. NOTE: If fields are not defined in the file, they cannot be specified in the entry port, and the entire line is searched. If fields are defined, they can be specified in the entry port, with only those fields being searched.
3	Repeat search pattern? • If yes, press <CTL> <F4> (forward), or <CTL> <F5> (backward). • If no, procedure is concluded.	Find command continues to search through displayed file from current cursor position, or Find command is discontinued.

Pattern and Field entry boxes

The screenshot shows a menu bar with the following items: FIND, FILTER, SORT, QUERY, OUTPUT, DEFINE CTRL FILES, OPEN, and CLOSE. The 'FIND' menu is currently open, displaying a sub-menu with the following options: Pattern, Field (optional), and Search Direction. The 'Pattern' option is selected, and a text entry box is shown below it. The 'Field (optional)' option is also shown with a text entry box below it. The 'Search Direction' option is shown with two buttons: 'Forward' and 'Backward'.

52948

Figure 14 Pattern and Field Entry Boxes Display Example

2.12 Filtering Data

Definition

The Filter command is used to display a subset of the data currently on the screen. After the condition is entered, and the filter state is turned on, only lines of data that meet the specified condition are shown. The entire file is not filtered when the command is entered; the data is filtered as it is displayed.

Filter state

The Filter command is in effect until the [OFF] target is selected. When the Filter state is set to “off,” the original unfiltered data is redisplayed.



ATTENTION

Fields must be defined in order to use the Filter command.

Procedure

The following procedure describes the steps required to use the Filter command.

Table 11 Filter Command

Step	Action	Result
1	With a file displayed on the screen, select the [FILTER] target on the command line.	Both Filter State targets and Condition entry box overlay the display. A sample Filter State and Condition entry box appears at the end of this procedure.
2	Select the [ON] target.	Target is highlighted.
3	Type in desired condition. Press <ENTER>. Reference: 2.20 Conditions for Query and Filter Commands	Records that meet the condition are displayed.
4	Were records found after condition was entered? • If yes, go to Step 5. • If no, select [FILTER] from the command line, and set Filter State to [Off], and press <ENTER>. NOTE: If the results of a filter generate more than one page of data, the message “filtering data” appears when you use the <PAGE FWD> key to view the next page of data.	You are returned to Documentation Tool initial display.
5	Save filtered data? • If yes, use the [OUTPUT] command. Reference: 2.14 Outputting Files for this command procedure. • If no, go to Step 6.	Filtered data is saved, or the Filter command is repeated.

Step	Action	Result
6	Repeat Filter command? • If yes, go to Step 1. • If no, go to Step 7.	Not applicable.
7	Select the [FILTER] target, then the [OFF] target, and press <ENTER>. NOTE: Filter state can remain on except when defining fields.	Filter command is turned off.

Filter State and Condition entry box

FIND FILTER SORT QUERY OUTPUT DEFINE CTL FILES OPEN CLOSE

Filter State

Condition

52949

Figure 15 Filter State and Condition Entry Box Display Example

2.13 Sorting Data

Definition

The Sort command arranges records within the data in ascending order.

Sort rules

The following rules apply to the sort command:

- Zero or more fields may be specified. If no fields are specified, the data is sorted according to the first field. If no fields are defined in the data, the entire record is treated as a field.
- If multiple field names are specified in the field name list, data is sorted by the first field. All entries equaling the first field are then sorted by the second field, continuing in this way for all fields specified.
- Values which are NaN (- - -) are listed at the end of the data listing.
- Numbers or date/time must be defined as either number or date/time field types in order to be sorted by value when using multiple fields. If the data is not sorted after pressing <ENTER>, the field type can be checked by using the <CTL> <F10> keys as described in 2.15 Defining, Changing, and Deleting Fields.
- If the error message “Sort contains bad data” appears, then data in fields did not convert to values. This can happen if date/time or number fields are in unrecognized formats. The records which did not convert go to the end of the data.
- When sorted data consists of several pages, the results of the sort reside in a temporary file. When sorted data consists of a small amount of data, the results are held in memory.

Sort example

The following tables demonstrate how the Sort command works when the file contains the fields: NODENUM, MODNUM, and SLOTNUM, and the fields are specified in the Field Name entry box. The Sort command sorts the data by the first field: in this case NODENUM, then MODNUM, and then SLOTNUM.

Table 12 Before Sort

NODENUM	MODNUM	SLOTNUM
7	1	2
5	2	1
5	1	2
5	1	1
7	1	1

Table 13 After Sort

NODENUM	MODNUM	SLOTNUM
5	1	1
5	1	2
5	2	1
7	1	1
7	1	2

Procedure

The following procedure describes the steps required to use the Sort command.

Table 14 Sort Command

Step	Action	Result
1	With file displayed on the screen, select the [SORT] target on the command line.	Both Sort State targets and Field Name entry box overlay the display. A sample Sort State and Field Name entry box appears at the end of this procedure.
2	Select the [ON] target.	The [ON] target is highlighted.
3	Sorting by specified fields? • If yes, type in field name(s) separated by commas or blanks, and press <ENTER>. • If no, press <ENTER>. NOTE: To abort Sort command, press <CTL> <BREAK>.	"Sort in progress" message appears on screen. When sort is complete, the "Operation complete" message appears with sorted data.
4	Save sorted data? • If yes see, 2.14 Outputting Files for this command procedure. • If no, go to Step 5.	Not applicable.
5	Continue with another sort? • If yes, go to Step 1. • If no, select the [SORT] target from the command line, then the [Off] target and press [ENTER]. NOTE: Sort state can remain on unless defining fields.	Data is returned to its original order.

NOTE

Defining the condition using the [SORT] target is not mandatory for a Journal. With R640 a new target, [JRNL SORT] is added. By enabling the JRNL SORT target, the results are sorted before display and not before retrieval. Disabling this target displays events in the order in which they were journaled.

Sort State and Field Name entry box
Figure 16 Sort State and Field Name Entry Box Display Example**Sorting Journal Queries**

With R640, a new target [JRNL SORT] is added in the [JOURNAL QUERY] page. By default this target is selected. The query results are sorted based on the [Date-Time] field.

When JRNL SORT is selected, results are sorted before displaying the results. If the target is not selected, then the events are displayed in the order in which they were journaled, which may or may not be in chronological order.

Sorting is more meaningful when multiple journals are queried. The results of a single journal are displayed in the chronological order.

Table 15 JRNL SORT Command

Step	Action	Result
1	Choose [Jrnl] option from the [QUERY][Build] menu.	A sample query page appears.
2	Enter the Start/End date and time in [Start Date-Time] and [End Date-Time] boxes.	The new Date and Time appears in the box.
3	Enter the Conditions and select journal type from the list under [Select one or more Journals].	The condition appears in the box and the journal types are highlighted.
4	Click the [JRNL SORT] and press <ENTER>. NOTE: By default [JRNL SORT] is selected.	Sorts the results before the results are displayed.
5	Disable [JRNL SORT] and press <ENTER>.	Displays the unsorted results of the new query.

07 Jun 02 10:40:343

FIND

FILTER

SORT

QUERY

OUTPUT

DEFINE CNTRL FILES

OPEN

CLOSE

Build

Select Pre-Built

Delete Pre-Built

HWY

UCN

Unit

Node

Jrnl

Save

Overwrite

JRNL SORT

Start Date-Time?

06/07/02 08:00:00

End Date-Time?

06/07/02 10:00:00

Retrieval Filter

Unit List?

Conditions:

Descriptor?

Select one or more Journals:

Process Alarms

Process Changes

Operator Messages

System Maint Messages

System Status Changes

System Error Messages

Status Notification

Sequence Of Events

Engr Change Messages

CTL CTL CTL CTL CTL CTL

F2

F4

F5

F8

F9

F10

U

D

R

L

T

B

DEL

FFWD

FBACK

PATH

ERRORS

FIELD

Figure 17 Journal Sort for Journal Query Setup

2.14 Outputting Files

Definition

The Output command is used to print, or save to a text file, the results of previous Documentation Tool operations such as queries and sorts.

With R640, the [Output in CSV] option is available to save the query results in CSV format. The file can be saved with a “.xx” extension as supported by the system.

NOTE

The [Output in CSV] option is supported for Journal queries and can be accessed from the [QUERY] page.

Keylock level

If you are sending output to file, the keyswitch must be in the ENG position. If you are sending output to a printer, the keyswitch can be in any position.

Save options

You have the following four save options when outputting to a file.

Table 16 Save Options

Option	Description	Comments
Overwrite with Field Definitions	Saves current data to the file, including field definitions.	DEFINE_FIELD directives are saved at the beginning of the file, so when the file is opened with the Open command, the fields are displayed. Example: .DEFINE_FIELD ENTITY 20 STRING SHOW 2 ENTITY is the field name generated by the query.
Overwrite without Field Definitions	Saves current data to the file, but does not save field definitions.	Usually done for raw data intended to be used in another context.
Append	Adds current data to an existing file.	• If current display has fields, then fields must match those already in file. • If current display does not have fields, then there should be no fields in the file to which you are appending.
Output in CSV	Outputs the data in CSV format	• Saves the output into a file in CSV format. • This output format is applicable only for a journal query.



ATTENTION

One of the four save options must be selected before you can output to a file. To create a new file, either the Overwrite with Field Definitions or the Overwrite without Field Definitions save option must be selected.

07 Jun 02 10:38:02 3											
FIND	FILTER	SORT	QUERY		OUTPUT		DEFINE CNTRL FILES		OPEN	CLOSE	
			U	ET							
To a File											
To a Printer			5:29	SY 74	ALM	NODE 31	NIM	UCN 07	ND 27	FILE_1	OK
Pathname net>temp>csv.xx											
Overwrite With Field Definitions											
Overwrite Without Field Definitions											
Append											
Output in CSV											
06/07/02 10:28:55 SY 74 ALM NODE 31 NIM UCN 07 ND 27 FILE_1 OK											
06/07/02 10:29:11 SY 76 NODE 03 INFORM REPORT (INFORM											
06/07/02 10:30:01 SY 74 RTN NODE 31 NIM UCN 07 ND 27 FILE_1 OK											
06/07/02 10:30:07 SY 74 ALM NODE 31 NIM UCN 07 ND 27 FILE_1 OK											
06/07/02 10:30:09 SY 76 NODE 03 INFORM REPORT (INFORM											
06/07/02 10:30:11 SY 76 NODE 03 INFORM REPORT (INFORM											
06/07/02 10:31:08 SY 74 RTN NODE 31 NIM UCN 07 ND 27 FILE_1 OK											
06/07/02 10:31:11 SY 76 NODE 03 INFORM REPORT (INFORM											
06/07/02 10:31:14 SY 74 ALM NODE 31 NIM UCN 07 ND 27 FILE_1 OK											
06/07/02 10:32:10 SY 76 NODE 03 INFORM REPORT (INFORM											
06/07/02 10:32:12 SY 76 NODE 03 INFORM REPORT (INFORM											
06/07/02 10:32:20 SY 74 RTN NODE 31 NIM UCN 07 ND 27 FILE_1 OK											
06/07/02 10:32:27 SY 74 ALM NODE 31 NIM UCN 07 ND 27 FILE_1 OK											
CTL	CTL	CTL	CTL	CTL	CTL	F2	F4	F5	F8	F9	F10
U	D	R	L	T	B	DEL	FFWD	FBACK	PATH	ERRORS	FIELD

Figure 18 Journal Query with Output in CSV format Example

Procedure

The following procedure describes the steps required to use the Output command.

Table 17 Output Command

Step	Action	Result
1	With the file or query display currently on screen, select the [OUTPUT] target on the command line.	Targets appear for specifying destination: [TO A FILE] [TO A PRINTER]
2	Outputting to a file? - If yes, select the [To a File] target. - If no, go to Step 4.	Entry box for file pathname and the save options targets appear. A sample Output command pathname and save options figure appears at the end of this procedure.
3	Type in file pathname, select appropriate save option, and press <ENTER>. NOTE: Reference the table in: Save options.	"Request in progress" message appears, then "Operation complete." Display data is saved to the named file.
4	Outputting to a printer? - If yes, select the [To a Printer] target, type printer ID in entry box (for example, \$P4), and press <ENTER>. - If no, procedure is concluded.	"Request in progress" message appears, and data is printed. "Operation complete" message is displayed. NOTE: Virtual printers (VP) are illegal. Direct output to a real printer only.

Output command pathname and save options

DD MMM YY HH:MM:SS 1

FIND	FILTER	SORT	QUERY	OUTPUT	DEFINE CNTRL FILES	OPEN	CLOSE		
STATEMENT									
SEND (WAIT):"READY THE REACTOR"									
SEND (WAIT):"ENTER NEW TIC21000 SETPOINT (RANGE= 0_100 DEGC)"									
SEND : "MIXTIME WILL BE",AGTIM000.SP," SECONDS"									
SEND : "BATCH COMPLETE"									
SEND (WAIT):"TANK_A VALVE FAILED: REPAIR AND CONFIRM"									
SEND :PROGRAM RESTART AT PHASE LOADA"									
SEND (WAIT):"TANK_B VALVE FAILED: REPAIR AND CONFIRM"									
SEND : "BATCH RESUMES AT PHASE LOADB"									
SEND : "Batch not satisfactory; recycle product."									
SEND : "EXCESS SOLUTION",EXCESS,"GALLONS"									
CTL U	CTL D	CTL R	CTL L	CTL T	CTL B	F2 DEL	F4 FFWD	F5 FBACK	F8 F9 F10 PATH ERRORS FIELD

5294

52941

Figure 19 Output Command Pathname and Save Options

2.15 Defining, Changing, and Deleting Fields

Procedure

The following procedures describe how to define, change, and delete fields.

Table 18 Defining Fields

Step	Action	Result
1	With file displayed on the screen, move the cursor to the last character to be included in the field, and press <CTL> <F10>. NOTE: Fields cannot be defined if the Filter or Sort state is "On."	A menu appears, prompting for the field name and type. If cursor is not on last column of existing field, then a new field is defined from end of preceding field, up to and including the cursor position. A sample Field name and Type menu appears at the end of this procedure.
2	Type in field name, select type, and press <ENTER>. NOTE: Sixteen characters are allowed for the field name, but only displays characters that fit the width of the field.	The field boundary is defined and indicated by a vertical line to the right of the cursor.
3	Save newly defined field(s)? • If yes, see 2.14 Outputting Files for this command procedure. • If no, new fields disappear when file is closed.	This is done if you need to use this data for future reference.

Table 19 Changing Fields

Step	Action	Result
1	Move the cursor to the last character in the field and press <CTL> <F10>. Type in the new field name, and press <ENTER>. NOTE: Field names created as a result of a query cannot be changed.	A menu appears, prompting for the new field type. A sample Field name and Type menu appears at the end of this procedure.
2	Change type of an existing field? • If yes, move the cursor to the last character in the field and press <CTL> <F10>. Select the new field type, and press <ENTER>. • If no, field type(s) remain unchanged.	A menu appears, prompting for the new field type. A sample Field name and Type menu appears at the end of this procedure.

Table 20 Deleting Fields

Step	Action	Result
1	Delete existing fields? - If yes, move cursor to the last character in the field to be deleted, and press <CTL> <F10>. Field name and type are displayed. Select [Delete Field] target, and press <ENTER>. - If no, procedure is concluded.	The field name and field definition indicated by the blue line disappears.

Field name and Type menu

Field Name

Field Type

52952

Figure 20 Field Name and Type Menu Display Example

2.16 Building, Saving, Executing and Recalling a Query

Finding query- building information

Before building a query, you must be knowledgeable of the process network numbers, device numbers, node numbers, unit IDs, entity names, or entity types for your system. If you do not know these names and numbers, you can find them through the System Menu or the System Status display.

To learn which unit IDs are on your system, call up the System Menu, and select [ORGANIZATIONAL SUMMARY], [UNIT TITLES], then select [DISPLAY]. The Unit Titles display appears listing all unit IDs and unit names.

To determine your process network numbers, call up the System Status display. The process network targets, along with the network numbers appear in the lower half of this display.

To determine entity types, refer to the Event Types table which follows.

Saving query data

There are two types of query data that you may save:

1. The setup data that is entered before the query is executed.
2. The results data that is displayed after the query is executed.

Saving query for Recall

With R680, the last executed query is automatically saved to the SYSQRY.DC file. The recall operation uses this file to retrieve the last executed query. When the query is executed successfully and the save operation fails, an error message is displayed.

The saved query is not affected by the station reload and is available even after exiting the Doc tool.



ATTENTION

Even if there is a run-time error, the query is saved. However, if there is a run-time error and also a failure in save operation, only the run-time error message is displayed.

Saving setup data

Setup data for a query can be saved at any time during the [BUILD] process by typing in a query descriptor and selecting the [SAVE] target.

After the setup data is saved, you can choose [SELECT PRE-BUILT], modify the data, and save it again under the same or a different descriptor. When modifying an existing (prebuilt) query, you use the [OVERWRITE] target.



ATTENTION

Remember to select the Save or Overwrite target before pressing <ENTER> to execute the query. After a query is executed, the setup data is cleared and, if you have not previously saved it, the setup data cannot be retrieved.

Saving results data

You save the results of a query before cancelling the display by using the Output command. Results can be saved to a printer or a user-named file on storage media.

Recalling a query

With R680, you can recall the last executed query. The query is automatically saved, when the field entries are correct. If there is a mismatch, an error message is displayed. The run time error does not have any impact on the automatic saving of a query. When a query is executed, the name in the Descriptor port is ignored and the query is saved with a predefined query name.



ATTENTION

If you execute a query with some or all blank fields, the last saved query is overwritten.

When you choose [RECALL LAST QUERY], the last executed query appears. In the last executed query, all the query parameters appear, with the query Descriptor port blank.

After the execution, the query on each station is saved with a reference to the station number. The recall operation calls up the last executed query on that station. The last executed query for each station is not listed in the pre-built query list.

Before beginning

If you are using removable media with the Documentation Tool, perform the following procedure before saving or executing queries:

Table 21 Removable Media Step

Step	Action
1	Call up the Modify Default Volume Path Name display, and enter pathnames for the following directories: • Documentation control directory (DOC CTL DIR) • Temporary file directory (TEMP FILE DIR)

Modify Default Volume Path Name display

DD:MM:YY HH:MM:SS 1

MODIFY DEFAULT VOLUME PATH NAMES

Edit all desired default paths and ENTER

HG GDF [NET>&HGG>]	NETWORK CONFIG [NET>&ASY>]	CL OVERLAY [NET>&OP2>]	DEB OVERLAY [NET>&OP1>]	SDT OVERLAY [NET>&OP4>]
HM/AM/CM GDF [NET>&AMG>]	CL SOURCE/OBJ [NET>CL>]	NOT USED [NET>&OP1>]	LBC OVERLAY [NET>&OP1>]	FIND NAMES OVLY [NET>&OP4>]
AREA DB GDF [NET>&ARG>]	CL PARAM LIST [NET>CL>]	NOT USED [NET>&OP1>]	TRANSLATORS OVL [NET>&OP4>]	LOAD NODE OVRLY [NET>&OP4>]
CL CUSTOM GDF [NET>&CDSG>]	USER DEFLT PATH [NET>TEST>]	BUTTN CFG OVRLY [NET>&OP1>]	CONFIG OVRLY [NET>&OP1>]	PICTURE EDITOR [NET>&OP2>]
NIM GDF [NET>&NMG>]	KEY FILE VOLUME [NET>&KFO>]	SMCC OVERLAY [NET>&OP4>]	TAC SUPPORT OVL [NET>&OP5>]	GENERIC OVERLAYS [NET>&OVG>]
NIM GDF [NET>&NM2>]	EXT LOAD MODULE [NET>&CUS>]	DOC CTL DIR [NET>&DOC>]	TEMP FILE DIR [NET>THL>]	NCF BACKUP PATH [NET>&OP4>]
SET DEVICE PATH TO REM. MEDIA	SET DEVICE PATH TO "NET"	MAIN MENU	UTILITIES MENU	

52953

Pathname for Documentation Control directory

Pathname for Temporary File directory

Figure 21 Modify Default Volume Path Name Display Example

Procedure

Use the following procedure to build, save, and execute a query.

Table 22 Build, Save, and Execute a Query

Step	Action	Result
1	Select the [QUERY] target from the command line.	Targets appear for the 4 query operations: [BUILD] [SELECT PRE-BUILT] [DELETE PRE-BUILT] [RECALL LAST QUERY] (R680)
2	Select the [BUILD] target.	Targets appear for the database types that can be queried: [HWY] [UCN] [UNIT] [NODE] [JRNL] [FNM]
3	Select appropriate target for desired database: - To Access data from an entire hiway, a list of hiways, a single box, or a list of boxes on one or more hiways, select [Hwy]. - To access data from an entire UCN, a list of UCNs, a single UCN device, or a list of devices on one or more UCNs, select [UCN]. - To access data from a single unit or a list of units, select [Unit]. - To access data from a single LCN node or a list of LCN nodes, select [Node]. - To access data from one or more journals, select [JRNL]. - To access a data form to build the Find Names query, select [FNM].	A query setup display appears. Each display varies slightly depending on the database you have selected. A sample Query display appears at the end of this procedure. Reference >> Query Journal Events for more information about this function for the R500 and later Documentation Tool. Reference >> Query Find Names (Find Point Interconnections) for more information about this function for the R660 and later versions of Documentation Tool.
4	Type in the query setup data. WARNING: Do not press <ENTER> yet! Reference the table found in: 2.18 Rules for Entering Query Setup Data.	Not applicable. NOTE: If you select Jrnl query, when the setup display appears you can press the <ENTER> key so that default date-time values appear for the preceding 24-hour period.

Step	Action	Result
5	Save query “setup” data? • If no, go to Step 6. • If yes, type in descriptor and select the [Save] target.	Query is saved to SYSQRY.DC file in the DOC CTL directory, and is now available as a prebuilt query. If an error message appeared, see 2.26 Setting the Directory Pathname.
6	Execute the query now? • If no, press <CANCEL> key to clear display and conclude procedure. • If yes, press <ENTER> key. NOTE: <CTL> <BREAK> can be used to abort a query.	Query results appear on the screen or If no data was found to match specified criteria, the message “No Match Found” appears. NOTE: When the message “No match Found” appears the descriptor is overwritten with blank. If any entries were invalid, an error message appears and the entry changes to red. NOTE: To manipulate query results, reference: 2.11 Finding Data 2.12 Filtering Data 2.13 Sorting Data 2.15 Defining, Changing, and Deleting Fields
7	Save query results? • If no, procedure is concluded. • If yes, select the [OUTPUT] target on command line. Reference: 2.14 Outputting Files	Not applicable.
8	Select the [ReCALL LAST QUERY] target (R680 and later).	The last executed query screen appears.
9	Modify the values if needed and execute the query again.	Query results appear on the screen.

Query display

11 Mar 05 11:30:45 3

FIND	FILTER	SORT	QUERY	OUTPUT	DEFINE CNTRL FILES	OPEN	CLOSE
------	--------	------	-------	--------	--------------------	------	-------

Build	HWY	JCN	
Select Pre-Built	Unit		
Delete Pre-Built	Node		
	Jrnl		
	Fnm		

Save

Overwrite

Proc Net List?

Device List?

Entity Names?

Conditions:

Param values to show:

Descriptor?

Resource/Entity Types:

ALL

DIG IN

DIG OUT

DIG COMP

ANL IN

ANL OUT

LOGIC

DEV CTL

REG PV

REG CTL

FLAG

TIMER

NUMERIC

ARRAY

PROC MOD

CM

CTL	CTL	CTL	CTL	CTL	CTL	F2	F4	F5	F8	F9	F10
U	D	R	L	T	B	DEL	FFWD	FBACK	PATH	ERRORS	FIELD

Figure 22 Query Display Example

Query result

QUERY				
ENTITY	PV	SP	MODE	
FIC101	50.5	50.8	AUTO	
FIC102	55.2	59.1	CAS	
FIC103	100.0	100.0	AUTO	
FIC202	25.4	30.5	CAS	
FIC204	33.4	52.2	MAN	
FIC301	24.4	21.3	MAN	
FIC303	75.6	75.5	AUTO	
FIC401	-----	0.0	MAN	
FIC404	56.5	56.6	CAS	

52946

Figure 23 Query Result Example

Query Journal Events

Overview

For R500 and later releases, Documentation Tool has a Query function Journal (Jrnl) option for accessing, filtering, and sorting data from real- time journals.

You can use the F10 key to redefine the fields, and then sort on this redefinition.

Press the <CANCEL> key to exit any of the Documentation Tool targets.

Select the Output target to save the resulting data to a file or to a printer.

To clear the Query results without saving them, press the <CANCEL> key.

With R640, the following enhancements are added to the journal query page to improve the performance of the Documentation Tool:

- JRNL SORT sorts the records before displaying the records.
- Retrieval Filter filters the records before the retrieval phase and displays only the relevant events.
- Query result output is saved in a CSV format.

NOTE

The file extension should not be more than two characters.

- Two special symbols (@ and ~) represent the date and time for the journal query.



ATTENTION

The R640 enhancements are applicable to Journal queries. These are not applicable to any non-journal queries like HWY, UCN, Unit and Node. The Background queries (using CL, CP, and Schematic actors) are not affected by the R640 enhancements.

Comparing Query Journal to Event History

This Query Journal option differs from Event History in several ways:

- Journal allows you to query any or all of the journals existing on the system pertaining to the process units specified by the user in the unit list as shown in the figure that follows. Event History only allows you to select one journal at a time, however it may present four or five pages of events for a journal.
- Journal allows you to do chronological sorting of events across units and /or report types.
- Journal allows you to further filter or sort and then save data that has been collected to a file or printer. Event History data can also be saved to a file using the virtual printer technique.

- Journal allows you to see Sequence of Events (SOEs) that reside on multiple History Modules and to operate on this data—filter, sort, and so on.

Query Journal setup display

The following figure represents the Query Journal setup display, with sample data parameters.

Figure 24 Query Journal Setup Display Example

Retrieval Filter for Journal Queries

With R640, the process of defining the filter state for a Journal query is slightly different from other queries.

In normal circumstances, the Journal query events are being filtered based on the specified condition during the display stage.

By using the [Retrieval Filter] option provided in the query page, the filtering of events can be done at the retrieval stage itself. By default, this option is disabled. Enter the condition and enable the [Retrieval Filter] option to filter the events during the retrieval stage. This results in displaying only the events that are relevant to the specified condition. Since the number of records becomes less due to filtering, it leads to a reduction in overall execution time.

Table 23 Retrieval Filter Command

Step	Action	Result
1	Choose Jrnl option on the QUERY → Build menu to display the query page of the Journal Query.	A sample query page appears at the end of this process.
2	Specify the Start/End date and time, Condition, and Journal Type to retrieve the query results, as required.	Date and Time, and Condition appear on the respective field. The Journal types are highlighted.
3	Enable the [Retrieval Filter] option and press Enter.	Filters the events during their retrieval stage and displays the result.
4	Disable the [Retrieval Filter] option.	Filter condition will be applied during the display stage and filters the events.

07 Jun 02 10:43:24 3									
FIND	FILTER	SORT	QUERY	OUTPUT	DEFINE CNTRL FILES	OPEN	CLOSE		
Build		HWY							
Select Pre-Built		UCN							
Delete Pre-Built		Unit							
		Node							
		Jrnl							
				Save	Overwrite	JRNL SORT			
Start Date-Time?	@ - 30								
End Date-Time?	^ - 10	Retrieval Filter							
Unit List?									
Conditions:	ET = 76								
Descriptor?									
Select one or more Journals:									
Process Alarms			Process Changes			Operator Messages			
System Maint Messages			System Status Changes			System Error Messages			
Status Notification			Sequence Of Events			Engr Change Messages			
CTL	CTL	CTL	CTL	CTL	CTL	F2	F4	F5	F8 F9 F10
U	D	R	L	T	B	DEL	FFWD	FBACK	PATH ERRORS FIELD

Figure 25 Retrieval Filter for Journal Query Example

NOTE

Defining the filter state from the FILTER main menu for a Journal is not mandatory, because there is a Retrieval Filter option provided for the user in the query page itself. The option can be enabled to filter the events at the retrieval stage. Disabling this option will filter the events during the display stage.

Query Journal results

The following figure represents the query results from selected journals on the Query Journal display.

FIND	FILTER	SORT	QUERY	OUTPUT	DEFINE CNTRL FILES	OPEN	CLOSE
DT			U	ET	GI		
09/10/96 10:06:00		SY	01	US04	\$\$NODE_ADMIN	SOFTWARE	
09/10/96 12:08:00		SY	02	US04	\$\$WATCHDOG	COMMUNICATN	
09/10/96 14:15:00		SY	01	US01	E\$\$_SDB_TASK	SOFTWARE	
09/10/96 15:03:00		SY	01	US10	\$\$LOAD_DUMP	SOFTWARE	

54153

Figure 26 Query Journal Results Example

Query Find Names

Overview

With R660, the Find Names (Fnm) query option (otherwise known as “Finding Point Interconnections”) has been added to the Documentation Tool. A feature of the point interconnection function is that it searches the point data checkpoints in a timely manner.

The Fnm query option allows the engineer to build pre-defined queries that can search for point interconnections in AM, UCN, and Data Hiway checkpoints, including AM/CL programs, and xPM and MC CL sequences. The Documentation Tool Fnm query is built and saved with the descriptor entry in the SYSQRY.DC file. The user can not execute this query within the Documentation Tool.

Using the predefined queries with the \$INTRCON schematic, an operator can find data points that are interconnected with Application Module, Universal Control Network, and Hiway Gateway data points and associated CL programs and sequences in the control strategy.

The \$INTERCON schematic is an electronic form that allows the user to enter the name of the Documentation Tool query which defines the limits of the search. The \$INTRCON schematic allows the user to start the search and view the results of the search.

Refer to the *Engineer's Reference Manual*, section 37, "Finding Point Interconnections". The Finding Point Interconnections function is similar to the Find Names function available through the Command Processor in that they both search for point interconnections in checkpoint files. However, the Command Processor Find Names function has an engineer keylock, and the Finding Point Interconnections function has an operator keylock. The Find Names function has many more options than the Finding Point Interconnections function. In Finding Point Interconnections, the search is limited to checkpoint database searches for entity references to points and CL programs.

Query Build Find Names Form Display

The following figure shows the Query Build Find Names Form display.

08 Mar 05 11:09:47 8									
FIND	FILTER	SORT	QUERY	OUTPUT	DEFINE	CNTRL	FILES	OPEN	CLOSE
Build		HWY							
Select Pre-Built		UCN							
Delete Pre-Built		Unit							
		Node							
		Jrnl							
		Enm							
					Save	Overwrite			
AM Unit List?									
AM CKPT Path?									
UCN List?									
UCN Node List?									
UCN CKPT Path?									
HG List?									
HG CKPT Path?									
CL?	AMCL	UCNCL	HGCL						
Printer?									
File Out Path?									
Descriptor?									
Note: Not a Documentation Tool Function									
Used by operator to search for selected point									
CTL	CTL	CTL	CTL	CTL	CTL	F2	F4	F5	F8
U	D	R	L	T	B	DEL	FFWD	FBACK	PATH
									F9
									ERRORS
									F10
									FIELD

Figure 27 Query Build Find Names Form

Query Build Form for Point Interconnection Details**Procedure**

The following table lists and defines the rules for entering Operator Find Name query data into the Documentation Tool. The entry column corresponds to ports and targets on the Find Names query displays.

Table 24 Search Rules for Entering Query Find Name Data

Entry	Definition	Rules
AM Unit List	Search the specified AM unit checkpoint	• Leave blank when the AM unit list search is not needed. • Enter an asterisk (*) (to search all AM unit checkpoints. • When listing more than one AM unit, insert a blank or a comma (,) between each AM unit number.
AM CKPT Path	Search the AM checkpoint file path.	• Leave blank to search the AM checkpoint on the NET (History Module). • Specify the path as \$Fn>* if the AM checkpoints are on removable media.
UCN List	Search a specific UCN	• Leave blank when the UCN search is not needed. • Type an asterisk (*) to search all UCNs in the system. • When listing more than one UCN number, insert a blank or a comma (,) between each UCN number.
UCN Node List	Specify the UCN node checkpoints to be searched on the specified UCN.	• Type an asterisk (*) or leave blank to search all UCN node checkpoints for the specified UCN. • When listing more than one UCN node number, insert a blank or a comma (,) between each UCN node number.
UCN CKPT Path	Specify the UCN checkpoint file path.	• Leave blank to search the NIM/xPM checkpoints and xPM/CL objects on the NET (History Module). • Specify the path as \$Fn>* if the UCN checkpoints and objects are on removable media.
HG List	Specify the HG node checkpoints to be searched on this LCN.	• Leave blank when the HG search is not needed. • Type * to search all HG checkpoints on this LCN . • When listing more than one HG node number, insert a blank or a comma (,) between each HG node number.

Entry	Definition	Rules
HG CKPT Path	Specify the HG checkpoint file path.	• Leave blank to search the HG checkpoints and MC CL objects on the NET (History Module). • Specify the path as \$Fn>* if the UCN checkpoints and objects are on removable media.
CL?	AM CL target	• Select AM CL target to search the AM CL in the checkpoints specified by unit numbers the AM unit list.
	UCN CL target	• Select UCN CL target to search the UCN List and UCN Node List in the specified UCN CKPT path.
	HG CL target	• Select HG CL target to search the Multifunction Controller (MC) files in the specified HG CKPT path.
Printer	Specifiy the printer number that is to print the results of the interconnection search.	• Leave blank if you do not want to provide the operator the ability to print the results of the point interconnection search. • Type the printer number as \$Pn (console printer number) to provide the operator an option to print the results of point interconnection search.
File Out Path	Specify the pathname of the file where inteconnection search results are to be saved.	• Leave blank if you do not want to provide the operator an option to write the results of the point interconnection search into a file. • For example, type the file pathname as NET>TEST>FileName.xx to provide the operator an option to save the results into a file. NOTE: .xx is the required extension for this file.
Descriptor	Specify the name of the query.	• Type the name of the form so that the operator can use this name to execute a point interconnection search. The descriptor can have a maximum of 64 characters. NOTE: The descriptor name should not be "--" or a "NULL" string. These strings are interpreted by \$INTRCON schematic support as an empty string and the parameter values are cleared.

NOTE

The saved query can be used only by the operator with the \$INTRCON form for a point interconnection search, and cannot be used within the Documentation Tool.

Results of a point interconnections search using an FNM query descriptor

In the following example, the engineer has shortened the search of all AM/HG/UCN databases to one AM unit of concern. The engineer has configured an FNM Query descriptor named “AM unit B1 only without print and file” to shorten the search time for the operator. The engineer then tells the operator to search for all point interconnections for point AMTREND11.

(To implement the point interconnection function at the station where the operator is located, the engineer must have previously loaded the OPFNAM and/or UPFNAM external load modules into the station.)

One of the ways for the operator to initiate a point-interconnection search for the AMTREND11 point is to select the point name AMTREND11 from a group display, press the SCHEM key, and then enter \$INTRCON as the name of the schematic. When the \$INTRCON schematic appears as shown in Figure 28, the AMTREND11 point name appears in the ENTITY port. Then, the operator enters “AM unit B1 only without print and file” into the FNM Query port, elects the Request target, and presses ENTER to initiate the search for point interconnections. When the search is completed, Request returns to OFF and the Status returns to INACT.

As shown in Figure 28, the bottom portion of the \$INTRCON schematic displays the complete search results. The result shows that AM unit B1 contains three references to point AMTREND11 parameters. The PVHITP parameter is referenced by the AMTREND12 PV input connection. In addition, the CL AMCLB linked to point AMCUS has references to the PVLOTP, PVHITP, DEVLOTP, PV, and SPP parameters of point AMTREND11 within CL SET statements.

11 Mar 05 11:31:02 3

NETWRK VIEW <>

\$INTRCON - Connection Search Schematic for 'Selected Point' AMTREND11

FNM QUERY

Status INACTIVE Entity AMTREND11

PARAM...	CL.....	ENTITY.....	PARAM....	CL.....	ENTITY.....
01PVHITP		AMTREND12			
02PVLOTP	AMCLB	AMCUS			
03PVHITP	AMCLB	AMCUS			
04DEVLOTP	AMCLB	AMCUS			
05PV	AMCLB	AMCUS			
06SPP	AMCLB	AMCUS			

start time 11:29:47 end_time 11:30:14

Figure 28 Example of Interconnection Result

Examples of setting up pre-built FNM queries

The following are examples of setting up queries for Finding Point Interconnections:

Scenario 1

The operator has to initiate a point interconnection search that applies to the AM unit 8 checkpoint, including CLs on the NET, but is not allowed to print the search results nor send the search results to a file.

To set up a pre-built Fnm query that implements the requirements of this scenario, perform the procedure in Table 25.

Table 25 Defining Point Interconnection Search Parameters, Scenario 1

Step	Action
1	From the Documentation Tool main menu, select the QUERY target.
2	Select the Build target and the Fnm target on the Query Build form to display the Query Build Find Names Form.
3	Type 8 in the AM Unit List port.
4	Select the AM CL target.

Step	Action
5	Leave the AM CKPT Path, UCN List, UCN Node List, UCN CL, UCN CKPT Path, HG List, HG CKPT path, Printer, and File Out Path ports blank to disable these features.
6	In the Descriptor port, type “AM Unit 8 only without Print and File.”
7	Select the [SAVE] target.

Scenario 2

The operator has to initiate a point interconnection search that applies to all AM/UCN node checkpoints including CLs on the LCN, and is allowed to print the search results and send the search results to a file.

To set up a pre-built FNM query that implements the requirements of this scenario, perform the procedure in Table 26.

Table 26 Defining Interconnection Search Parameters, Scenario 2

Step	Action
1	From the Documentation Tool main menu, select the QUERY target.
2	Select the Build target and the Fnm target on the Query Build form to display the Query Build Find Names form.
3	In the AM Unit List, UCN List, and UCN Node List, type an * (asterisk) to search all AM and UCN checkpoints.
4	Select the AM CL and UCN CL targets.
5	In the Printer port type \$P1, and in the File Out Path port type NET>TEST>ALLTPN.XX where TEST is the your designated user volume or directory and ALLTPN is the filename supplied by you. NOTE: Extension XX is required in the file name.
6	Leave the AM CKPT Path and UCN CKPT Path ports blank to disable these features.
7	In the Descriptor port, type “All of TPN with print and file.”
8	Select [SAVE].

Event Types

The following table describes the Event Types.

Table 27 Event Types

Event Type	Description
0	Fatal Driver Error or Crash Error
1	Non-fatal Driver Error or Software Error
2	Communication Error
3	CPU Driver Error
4	CRB Driver Error
5	FDS Driver Error
6	WDS Driver Error
7	PIC Driver Error
8	DHI Driver Error
9	CLH Driver Error
10	SIO Driver Error
11	Floating Point Error
12	Exception Error
13	CLK Error
14	SPC Error
15	BL Error
16	PNI Error
17	NGI Error
18	WSI Error
19	Peripheral Error Event
20	Peripheral Event Recovery
21	Process Alarm
22	Sequence Event
23	CL Message
24	Process Alarm Change Event
25	CL Message Confirm
26	Values
27	SOE
28	Point/Unit/Console Alarm Disable/Inhibit, Unit Assignment, Sort, Filter, Suppress Audible, Freeze
29	Operator Comment
30	Sequence Ack
31	Alarm Ack
32	CL Msg Acknowledge

Event Type	Description
33	SOE Ack
34	CL Msg Clear
35	Silence Alarm
36	LCN Event
37	Highway Event
38	Journal Box Event
39	Display Box Event
40	System Event Change
41	Off-process System Change Event
42	Operator-initiated LCN Change
43	Operator-initiated Hiway Action Event
44	Operator-initiated Box Change
45	Operator-initiated Time Change
46	State Change Notification (Physical Node State, Logical Node State, Personality Change)
47	Slot Event
48	&ASY Restore Event
49	Maintenance Recommendation
50	Corrective Action
51	Maintenance Action
52	Maintenance Clear
53	TAC Connect
54	TAC Disconnect
55	Maintenance Acknowledge
56	LCN Event Acknowledge
57	Hiway Acknowledge
58	Box Acknowledge
59	Peripheral Acknowledge
60	Cable Status
61	Lost Process
62	Lost Peripheral Event
63	Lost Hiway Event
64	Cable Status Acknowledge
65	Lost Process Event Acknowledge
66	Process Event Recovery
67	Lost Peripheral Acknowledge
68	Journal Overflow Operator Message
69	OPT (On Process Test) Error
71	Lost Hiway Acknowledge

Event Type	Description
72	Hiway Event Recovery
73	Current Hiway Status
74	IOP Event
75	Current HPN Status Event
76	Auxilliary Status Event
77	Auxilliary Status Acknowledge
78	Operator HPN UCN Change Event
79	NIM, PM, APM, HPM, LM, IOP, Cable Acknowledge
80	HM Error Handler
81	HM Checkpoint Control
82	HM Operator Message System Event
83	Physical Node Status Change
84	Engineer Changes

2.17 Rules for Entering Query Setup Data

List of rules

The following table lists rules for entering query setup data. The Entry column corresponds to ports and targets on the query setup displays.

Table 28 Rules for Entering Query Setup Data

Entry	Definition	Rules
Process Network List	One or more Data Hiway or Universal Control Network (UCN) numbers. This entry limits the search to the network(s) listed.	- Legal entries are numbers 1 through 20. - Type in one number (1), a list of numbers separated by blanks or spaces (1 2 3), or a range (1-3).
Start Date-Time (Optional)	Beginning date-time of journal records you are searching for.	DD/MM/YY hh/mm/ss, day/month/year hour/minute/sec.
End Date-Time (Optional)	Ending date-time of journal records you are searching for.	DD/MM/YY hh/mm/ss, day/month/year hour/minute/sec.
Device List (optional)	One or more Data Hiway box numbers or UCN node numbers. This entry limits the search to the process device(s) listed.	- Leave blank for all devices on the hiway or UCN. - Legal entries are hiway box numbers 5 through 63, or UCN node numbers 1 through 64. - Type in one device number (9), a list of numbers separated by blanks or spaces (9 13 17), or a range (9-17).
Unit List	A two character unit name. This entry limits the search to the unit(s) listed. The word "AREA" which signifies that journals are to be retrieved from all units in the local area. The word "AREA" can be used in conjunction with unit names as follows: AREA FE SA NO EA 05 The word "ALL" which signifies that journals are to be retrieved from all units.	- Legal entries for unit names are the two-character unit names that have been specified in the NCF during unit names configuration. - Type in one unit name, or a list of units separated by blanks or spaces (FE SA NO EA 05) or commas (FW,FE,03). - The word "AREA" either by itself or followed by a list of unit names. - The word "ALL."
Node List	One or more LCN node numbers. This entry limits the search to the LCN nodes listed.	- The legal entries are numbers 1 through 96. - Type in one node (40), a list of nodes separated by blanks or spaces (40 50 60), or a range (40-60).
Entity Names (optional)	One or more entity names, one or more entity list files, or a combination of one or more entities and one or more entity list files.	- Leave blank to specify all points. - Use commas or blanks to separate point names. - Do not use system-created entities (\$entities).

Entry	Definition	Rules
Conditions (optional)	One or more expressions defining which conditions to search for. Entities that meet the condition are listed in the query results.	Reference: 2.21 Conditions for Query and Filter Commands
Parameter Values to show (optional)	One or more parameter names. Parameters listed in this port are displayed in the query results along with the point name.	Use commas or blanks to separate parameter names. Do not use special characters. NOTE: See the table in Parameters occurring on every point, page 48 for a listing of those parameters that occur on every point for each node on the system.
Descriptor (optional)	The name you want to assign to the query for future reference.	Legal entry is up to 64 alphanumeric and/or special characters. This entry is required if query is to be saved.
Resource/ Entity Types	Targets to select entity types to search. Selecting entity types speeds up the query since a smaller database is searched.	Select one or more entity types. For unit and node queries, the entity types are not shown until the resource type is selected: AM, CM, UCN, Hiway, or ALL.
Select one or more Journals	Targets to select journal types to retrieve events from.	Select one or more journal types. Reference: >> Obtaining Process Histories: Logs, Reports, Journals section , Process Operations Manual.
[SAVE] (optional) [OVERWRITE] (optional)	Targets used to save the query setup data to the SYSQRY.DC file in the Documentation Control directory.	If a query with the current descriptor already exists, select both Overwrite and Save targets to overwrite existing query.

Parameters occurring on every point

The following parameters occur on every point for each node in the system. All other parameters apply only to specific nodes, and are listed in the *Parameter Reference Dictionary* for that particular node. You enter the parameters listed here in the “Param value to show” entry box.

Table 29 Parameters Occurring on Every Point

Parameter	Type	Definition
ENT_TYPE	Enumeration	The entity type of the point.
UNIT, UNITNAME	Enumeration	The two letter unit name. The Documentation Tool returns both parameters as two ASCII characters.
CRB	Integer	The LCN number the point is built on.
SUBSCRIPT	Integer	The entity type subscript. This parameter is only for subscripted entity types.
NODE_NO	Integer	The node number where the point resides.
NAME	String	The external point name. It is up to 16 characters in length if the LCN is local, and up to 19 characters if the LCN is external.
NODETYPE	Integer	The node type where the point resides: 0 - CG, 1 - HG, 2 - HM, 3 - AM, 4 - Universal Station, 8 - NIM, 9 - NG

Queries with Selection List

Prior to R620, the Entity Name port allowed a set of entity names to be entered and the query was limited to that set of names. The port size restricted the number of names that can be entered in the port.

With R620 and later software, the capability of specifying entity names in the Entity Names Port is enhanced. An option is available which allows a pathname of a Entity list file to be specified. The entity list file can contain any number of entity names and may be in HM or in removable media (\$F).

**ATTENTION**

The syntax is checked on all names in selection list. If the errors encountered are less than three, they will be displayed on bottom of screen. If three or more errors are encountered, the user is prompted to press F9 to view all of the errors.

In the Query Background mode, any errors encountered are appended to the results file.

For a HWY / UCN / Unit / Node Query command, a file name can be specified in the Entity port.

For example:

```
NET>ENTS>ENTITY.XX
```



ATTENTION

The file pathname specified in the Entity Port need not exist to save the query.

For a HWY / UCN / Unit / Node Query command, more than one pathname can be specified in the input port of Entity Names box.

For example:

```
$F5>SEL>ENTITIES.XX  
NET>SEL>ENT.XX
```

Any combination of the file pathname and entity name is allowed.

For example :

```
$F5>SEL>ENTITIES.XX  FAC*  
APMCL01 #NET>SEL>LIST.XX
```



ATTENTION

Invalid file pathnames in the Entity Names port will not halt the query execution.

File Format for Selection List

The file specified in the Entity Names port can have the entities in any of the formats shown below.

- a) One entity name per line.

```
RAMP000  
RAMP0001  
RAMP0002  
FAQ*  
CA?P*
```

- b) Many entities per line separated by a valid delimiter. A delimiter can be a blank space or a comma (',') or an end of the line character.

```
RAMP000 RAMP0001 RAMP0002 FAQ*  
CA?P* T?GS* RAMP0009 RAMP0012  
RAMP0000  
RAMP0006 RAMP0007  
RA?P*  
TAG*
```

- c) Leading / lagging blanks are allowed.
- d) All the valid wild cards are allowed as part of entity name.
- e) Duplicate entity names are allowed.
- f) Comments are not allowed. Anything found in the file is considered as an Entity Name.



ATTENTION

The characters '>' and '.' are invalid as part of entity name.

WARNING:

Minor impact on the performance is expected in terms of time taken to complete a query. This is due to :

1. Additional entities being fetched.
2. Number of files that are operated during run time.

The impact is in direct proportion to both the number of entities specified in the selection list and the number of file pathnames specified.

2.18 Modifying and Executing a Prebuilt Query

Procedure

A previously built and saved query can be executed by performing the following procedure.

Table 30 Execute a Built and Saved Query

Step	Action	Result
1	Select the [QUERY] target from the command line.	Targets appear for the query operations: [BUILD] [SELECT PRE-BUILT] [DELETE PRE-BUILT]
2	Select the [SELECT PRE-BUILT] target.	A menu appears listing descriptors of all prebuilt queries, with each descriptor being a separate target. A sample Prebuilt Query descriptors menu appears at the end of this procedure.
3	Select query to be executed.	The setup data appears for the selected query.
4	Do you want to modify the data? • If no, go to Step 6. • If yes, type in new data. WARNING: Do not press <ENTER> yet! Reference the table found in: 2.18 Rules for Entering Query Setup Data.	Not applicable.
5	Do you want to save the data? • If no, go to Step 6. • If yes, save with different descriptor?	The “setup” data is saved and is now available as a prebuilt query.
6	Execute query now? • If no, procedure is concluded. • If yes, press <ENTER> key. NOTE: Use <CTL> <BREAK> to abort query.	Results of query are displayed. To manipulate query results, see 2.11 Finding Data 2.12 Filtering Data 2.13 Sorting Data



ATTENTION

When a pre-built journal query is selected, that does not have a unit port query, the default for the unit port is ALL, that is the journals from all the units will be retrieved.

Prebuilt Query descriptors menu

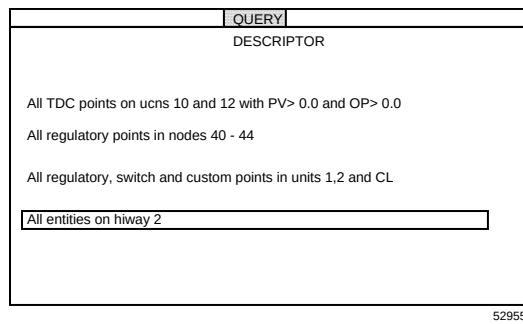


Figure 29 Prebuilt Query Descriptors Menu Display Example

For additional information, see the Pre-defined Documentation Tool Query section in the following documents:

- Actor's Manual
- Command Processor Manual
- Engineering Reference Manual

2.19 Deleting a Prebuilt Query

Procedure

A previously built and saved query can be deleted by performing the following procedure. The keylock level must be ENGR.

Table 31 Deleting a Built and Saved Query

Step	Action	Result
1	Select the [QUERY] target from the command line.	Targets appear for the query operations: [BUILD] [SELECT PRE-BUILT] [DELETE PRE-BUILT]
2	Select the [Delete Pre-built] target.	A menu appears listing descriptors of all prebuilt queries, with each descriptor being a separate target. Reference: Prebuilt Query descriptors menu
3	Position the cursor on the descriptor to be deleted and press <CTL> <F2>.	The query is deleted from the list of prebuilt query descriptors.
4	Save changes to the new list of pre-built queries? • If yes, select [CLOSE], and select the [SAVE] target. • If no, select or [CLOSE], and select the [DO NOT SAVE] target.	New prebuilt query list is saved with changes, or not saved with changes depending on which option is selected.

2.20 Conditions for Query and Filter Commands

Definition

A condition is composed of one or more expressions connected with logical operators. Expressions contain field names, comparison operators, and values. Both Filter and Query commands use conditions.



ATTENTION

When you use a condition with the Query command, you are entering parameter names. The parameter name does not have to be on the display. When you use a condition with the Filter command, you are entering field names. The field name must be on the display.

Release 610

For R610 and later releases, the following can be used on the right-hand side of the operator:

- **%parameter name** for HWY/UCN/UNIT/Node Query command
- **%field name** for Journal Query or Filter command

Condition format

Conditions and expressions are normally in the following formats:

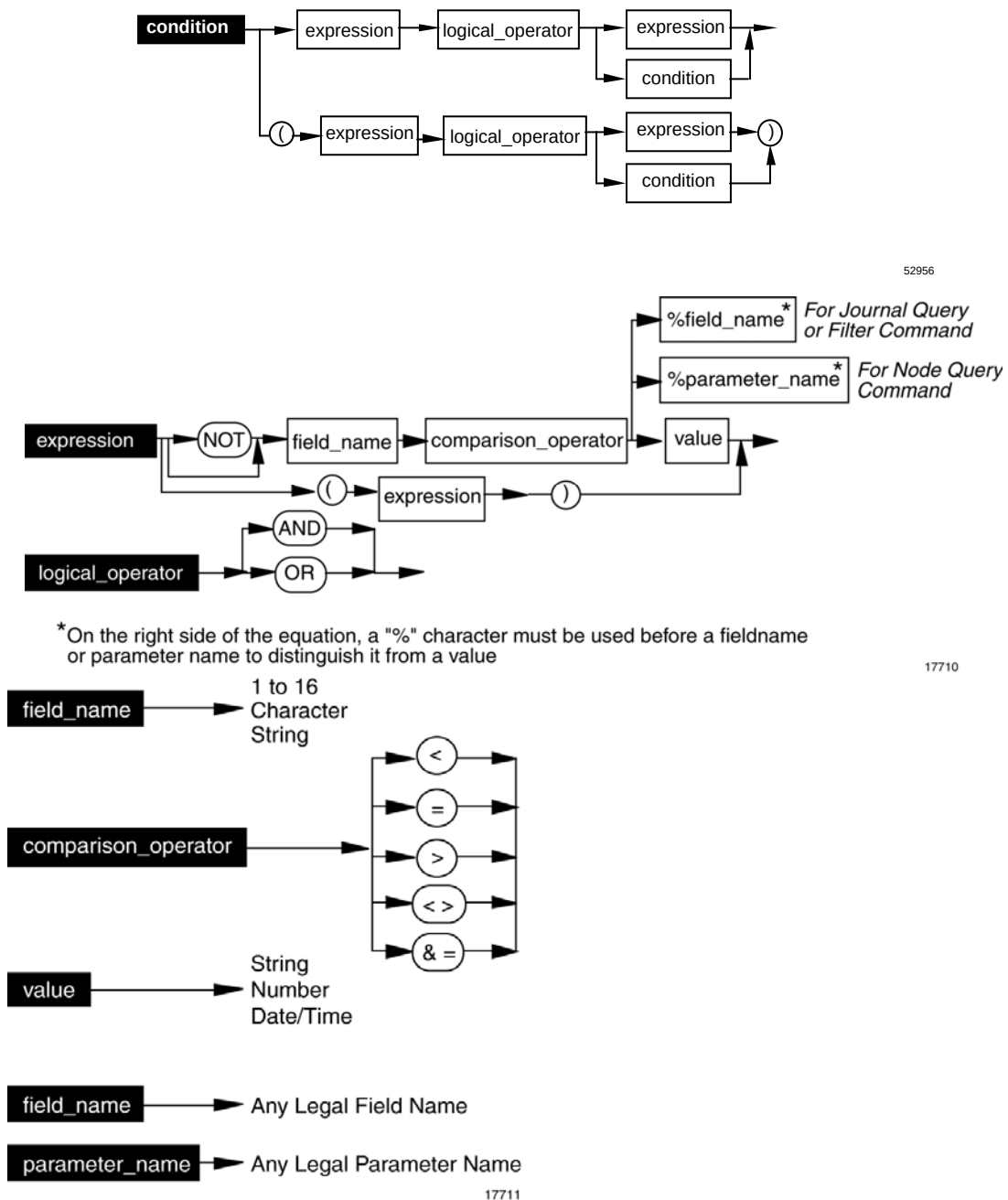


Figure 30 Conditions Format

Condition operator precedence

Operator precedence from highest to lowest is as follows:

2. ()
3. = <> < > &=
4. NOT
5. AND OR

The following table describes allowable condition operators.

Table 32 Condition Operators

Operator	Description	
Logical Operators	AND, OR	
Comparison Operators	<	less than
	>	greater than
	<>	not equal
	=	equal (must be an exact match)
	&=	contains (the data contains the string specified in the value)
()	May be used to order the evaluation, as in any arithmetic expression. Parentheses are not required.	

Condition with Filter command

(TAG = FIC21000) AND (TIME > 07:00:00)

where (continued)

TAG= Field name

== Comparison operator

FIC21000= Value

AND= Logical operator

TIME= Field name

>= Comparison operator

07:00:00= Value

Result:

Lines of data not meeting the condition are filtered (removed) from the display.

Condition with Filter command and fieldname on right side of operator

PV=%SP

Result:

All data lines where the process variable (PV) is equal to the setpoint (SP) will be shown.

Condition using [NOT] operator with Filter command

(TIME > 08:00:00) AND ([NOT] ALARM = BADPV)

where (continued)

TIME= Field name

>= Comparison operator

08:00:00= Value

AND= Logical operator

NOT= Not operator

ALARM= Field name

== Comparison operator

BADPV= Value

Result:

Lines of data whose alarm field is other than “BADPV” are displayed.

Condition with Query command

(HIWAY = 2) AND (BOXNUM > 5)
where (continued)
HIWAY= Field name
== Comparison operator
2= Value
AND= Logical operator
BOXNUM= Field name
>= Comparison operator
5= Value

Result:

Entity names that meet this condition are listed, i.e., the entity is assigned to hiway 2, and its box number is greater than 5.

Condition with Query command and parameter name on right side of operator

PV>70.0 OR MODE <>%NMode

Result:

All data lines where PV is greater than 70.0 or where Mode is NOT equal to its NMode will be shown.

Condition rules

The following table describes condition rules.

Table 33 Condition Rules

Item	Rule
Blanks	Blanks must be enclosed in single or double quotes. Example: DATE(1) > "01/02/91 12:00:00"
Special Characters (?, *, # or [])	Special characters are not allowed for field names, but are allowed for values that may be patterns. Example: (PTDESC = TANK*) OR (PTDESC = ?TNK[1-4])
Parameter Names	A parameter name can be used on either the left or right side of the operator. When it is used on the right side of the operator, it MUST be preceded by a percent "%" character.
Field Names	A field name can be used on either the left or right side of the operator. When it is used on the right side of the operator, it MUST be preceded by a percent "%" character.
Values	See the following table titled Values.

Table 34 Values

When the field type is...	Then...
A number and the operator is <,>	The value must be a number.
A number and the operator is &=, =, <>	The value may be a pattern (which may contain special characters or numbers).
A date/time and the operator is <, >	The value must be date/time.
A date/time and the operator is &=, < >, =	The value may be a pattern (which may contain special characters or date/time).
A string and the operator is =,<>,&=	The value may be a pattern (which may contain special characters).

Conditions for Query Journal

The following are your choices of pre-defined fields when entering conditions for a Query Journal (Jrnl):

- DT (date-time)
- ET (event type)
- GI (general information)
- U (unit)

For example, to query selected journals for entries of event type 27, you would enter the following in the Query Journal Condition port:

ET=27

2.21 Patterns for Query, Find, and Filter Commands

Definition

A pattern is a string the Documentation Tool matches using either the Find, Filter, or Query commands. A pattern can be used as a value in a condition.

Pattern special characters

The following special characters have meanings making it possible to specify a template, rather than an exact string.

Table 35 Special Characters to Use in Patterns

Special Character	Description	Result
?	Match a single character.	Example: A? Matches "A" followed by any character.
*	Match zero or more characters.	Example: *A Matches "A" preceded by any number of any characters.
[string]	Match any single character in <string>. Specifies a subset of characters that match. The characters specified between the brackets define what are considered a match. NOTE: Do not include ' ' inside the set unless it is one of the characters you want to match.	Example: A[BC]D Matches ABD or ACD.
[~string]	Not. Match any single character except those in <string>. A match occurs when it is not one of the characters specified in the set.	Example: A[~BC]D Matches AAD, ADD, AED, AFD.
-	A range of characters within the set to match. Used inside brackets.	Example: A[B-D]D Matches ABD, ACD, ADD
#	Match zero or more of the previous character.	Example: AB# Matches A, AB, ABB, AB BB
@	The special character that follows @ operates as the character itself.	Example: @? Matches only the "?" character.

2.22 Help Function

Overview

The Help function assists you in using Documentation Tool commands. All help information is contained in one text file, and you can use either the <PAGE BACK> or <PAGE FWD> keys to scroll through the file to locate the information you need, or the Find command to search for certain words in the file.

Accessing the Help function

You access the Help function by pressing the <HELP> key. The portion of the file that displays depends on what you have selected in the Documentation Tool.

Table 36 Help File Displays

If	Then
A command is selected	Help information applying to that particular command is displayed.
A command has not been selected	The beginning of the Help file is displayed.

Leaving the Help function

Once you have finished using the Help function, use the Close command or <CANCEL> key to close the display before proceeding with Documentation Tool commands.

Example

An example of a display in the Help file is shown in the following figure:

DD MMM YY HH:MM:SS 1

FIND	FILTER	SORT	QUERY	OUTPUT	DEFINE CNTRL FILES	OPEN	CLOSE
%230%				This function reads the current cursor position and puts up a menu which allows the user to enter:			
	F10 (mark field)			Field Name - Maximum of 16 characters Field type - String or Number or Date/Time Delete Field - Selected if the current field needs be deleted.			
				If the cursor position is not on the last column of an existing field, then a new field will be defined from the end of the preceding field up to and including the cursor position. If the columns are already included in a field, the size of that field will be reduced.			
				The user Opens the following file:			
				NET>TEST TESTA.XX 01/09/91 11:23			
				NET>TEST TEST1.XX 01/05/91 08:05			
				NET>TEST TESTB.XX 02/03/91 14:45			
CTL U	CTL D	CTL R	CTL L	CTL T	CTL B	F2 DEL	F4 FFWD
						F5 FBACK	F8 PATH
							F9 ERRORS
							F10 FIELD

52962

Figure 31 Help Display

2.23 Responding to Errors

How errors appear

Error messages appear on the current display in one of three ways:

- At the bottom of the screen describing why the current operation cannot be performed.
- As an error message box overlaying the current display, with a message describing why the operation was discontinued. These overlays appear when File Manager and out-of-memory errors occur.
- As a message directing the user to press the <CTL> <F9> (Errors) keys to display error messages. These keys are used when three or more errors occur during an operation.

Error box overlay

An example error message box is depicted here. Three targets are displayed in the error message box, and one of them must be selected before you can continue.

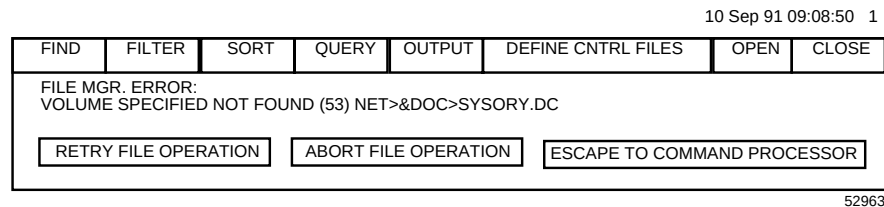


Figure 32 Error Box Overlay

Error box targets

A description of the error message box targets is as follows:

Table 37 Error Message Box Targets

Target	Description
[RETRY FILE OPERATION]	Reattempts the current Documentation Tool operation.
[ABORT FILE OPERATION]	Operation aborts and returns to prior screen of Documentation Tool operation.
[ESCAPE TO COMMAND PROCESSOR]	Calls up the Command Processor. Once in the Command Processor, you may perform operations such as setting the pathname for the Documentation Control directory, and deleting files if additional media space is required.

F9 error message

If you perform a function that creates more than two errors, an error message box does not appear. Instead, the message “Please use F9 to view the error” appears at the bottom of the current display. You then use the <CTL> <F9> keys to view the error display.

If you proceed with another function in the Documentation Tool, and do not call up the error display, it cannot be recalled. Once the error display has been called up, you can only remove it from the display stack by pressing the <CANCEL> key. It does not disappear when the next command is executed.

Display with F9 error message

DD MMM YY HH:MM:SS 1

FIND	FILTER	SORT	QUERY	OUTPUT	DEFINE CNTRL FILES	OPEN	CLOSE
ENTITY			CISRC(1)	CIDSTN(1)	CODSTN(1)	UNIT	
FIC 21000			AI21000.PV	PVAUTO	FPV21000.OP	01	
TIC21000			TI21000.PV	PVAUTO	FIC21000.SP	01	
FY21000			!!!	!!!	!!!	01	
FY22000			!!!	!!!	!!!	01	
FY23000			!!!	!!!	!!!	01	
FVL21000			!!!	!!!	!!!	01	
FVL22000			!!!	!!!	!!!	01	
DVL23000			!!!	!!!	!!!	01	
AG24000			!!!	!!!	!!!	01	
INGA000			!!!	!!!	!!!	01	
INGB000			!!!	!!!	!!!	01	
STATE000			!!!	!!!	!!!	01	
FULMT000			!!!	!!!	!!!	01	
CLNDT000			!!!	!!!	!!!	01	
AGTIM000			!!!	!!!	!!!	01	
SIMLT000			!!!	!!!	!!!	01	
REACT000			!!!	!!!	!!!	01	
SETNK000			!!!	!!!	!!!	01	
FVL24911			!!!	!!!	!!!	01	
FVL23911			!!!	!!!	!!!	01	
DVL23911			!!!	!!!	!!!	01	
PI23911			!!!	!!!	!!!	01	
Please use F9 to view the error.							

52964

Figure 33 Display with F9 Error Message

2.24 Out of Memory/Storage Space Errors

Ways to exceed memory or storage space

There are three ways you may exceed available memory or storage space when using the Documentation Tool:

- By performing a function that depletes Universal Station memory
- By exceeding the maximum number of files allowed for the directory as determined during the Create Directory command (CR)
- By performing a function that exceeds removable media storage space.

Exceeding Universal Station memory

When you run out of memory on the Universal Station, an error message box displays as follows.

Universal Station out of memory error box

DD MMM YY HH:MM:SS 1

FIND	FILTER	SORT	QUERY	OUTPUT	DEFINE CNTRL FILES	OPEN	CLOSE
DEPLETED RESOURCE							
Out of memory.							
CANCEL PRIOR DISPLAY				ABORT			
FVL22000	!!!	!!!	!!!	01			
DVL23000	!!!	!!!	!!!	01			
AG24000	!!!	!!!	!!!	01			
INGA000	!!!	!!!	!!!	01			
INGB000	!!!	!!!	!!!	01			
STATE000	!!!	!!!	!!!	01			
FULMT000	!!!	!!!	!!!	01			
CLNDT000	!!!	!!!	!!!	01			
AGTIM000	!!!	!!!	!!!	01			
SIMLT000	!!!	!!!	!!!	01			
REACT000	!!!	!!!	!!!	01			
SETNK000	!!!	!!!	!!!	01			
FVL24911	!!!	!!!	!!!	01			
FVL23911	!!!	!!!	!!!	01			
DVL23911	!!!	!!!	!!!	01			
PI23911	!!!	!!!	!!!	01			
CTL U	CTL D	CTL R	CTL L	CTL T	CTL B	F2 DEL	F4 FFWD F5 FBAC
						F8 PATH	F9 ERRORS
						F10 FIELD	

52965

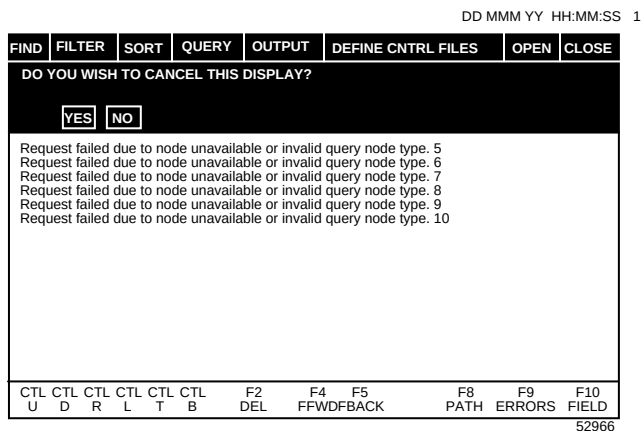
Figure 34 Universal Station Out Of Memory Error Box

Procedure

The following procedure describes the steps to take when you run out of Universal Station memory.

Table 38 Running Out of Universal Station Memory

Step	Action	Result
1	Select the [CANCEL PRIOR DISPLAY] target from the Out of Memory Error box.	The Cancel Error box displays. A sample Cancel Error box appears at the end of this procedure.
2	The display to be cancelled is shown. Do you wish to cancel it? • If yes, select the [YES] target. • If no, select the [NO] target.	The display is cancelled. You are returned to the prior display.
3	Select [close] target to close one or more displays.	Removes displays from display stack and creates more memory for the Universal Station. When a sufficient amount of memory is available, you can continue using the Documentation Tool.

Cancel Error box**Figure 35 Cancel Error Box****Exceeding storage space capacity**

When you run out of storage space on removable media, the error message box appears. The table found in Error box targets discusses how to use the targets shown in this error message box.

Removable media out of memory error box

DD MMM YY HH:MM:SS 1

FIND	FILTER	SORT	QUERY	OUTPUT	DEFINE CNTRL FILES	OPEN	CLOSE
FILE MGR. ERROR: INSUFFICIENT MEDIA STORAGE SPACE AVAILABLE (26) \$F2>PMC101461030_ _ _							
RETRY FILE OPERATION ABORT FILE OPERATION ESCAPE TO COMMAND PROCESSOR							
Device List?							
Entity Names?							
Conditions: OUTSSLT > 2							
Param values to show: OUTSSLT							
Descriptor? GENE'S 04							
Resource/Entity Types:							
ALL COUNTER DIG IN DIG OUT DIG COMP ANL IN ANL OUT ANL COMP							
FLAG TIMER NUMERIC LOGBLK REG CTL CNTR PROC MOD							
Query in progress.							

52967

Figure 36 Removable Media Out Of Memory Error Box

Deleting files

If you run out of storage on removable media, you can delete files to create additional space to continue using the Documentation Tool.

Procedure

Use the following procedure to delete files.

Table 39 Delete File

Step	Action	Result
1	Select the [ESCAPE TO COMMAND PROCESSOR] target on error message box.	Command Processor displays.
2	Enter the delete command and the name of the file to delete, or specify all (*.*) Example: DEL \$Fn>TEST>*.*	"DELETE COMPLETE" appears.

2.25 Setting the Directory Pathname

Why set the pathname?

The directory pathname must be set if the error message “VOLUME SPECIFIED NOT FOUND” appears in an error message box as depicted in the following. The message appears for one of the following reasons:

- The specified directory had not been created.
- The pathname to the Document Control directory had not been specified properly.
- The SYSQRY.DC file had not been created using the [DEFINE CNTRL FILES] target.

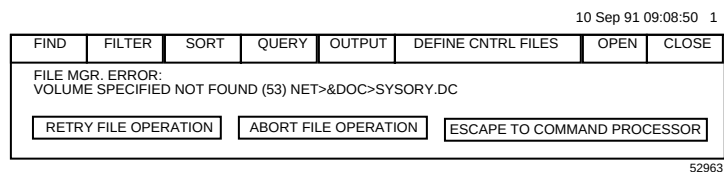


Figure 37 Volume Specified Not Found Error Message

Procedure

The following procedure describes the steps required to set the directory pathname.

Table 40 Setting the Directory Pathname

Step	Action	Result
1	Select the [ESCAPE TO COMMAND PROCESSOR] target on error message box.	Command Processor display appears.
2	Key in SP, and press <ENTER>.	Volume Pathname display appears.
3	Enter the directory names. NOTE: Default directory names are &DOC (Documentation Control), and TFIL (Temporary File), respectively. You may change these names.	Volume Pathname display is modified.
4	Select the [MAIN MENU] target on the Volume Pathname display.	Command Processor is displayed.
5	Press <CTL> <MENU>.	Error message box overlay in the Documentation Tool displays.
6	Select [RETRY FILE OPERATION] on the error message box overlay.	Error message box overlay disappears and previous Documentation Tool display appears.

2.26 Error Messages

Overview

The following error messages can appear while using the Documentation Tool. Some messages are informative, and do not require you to take any action, while other messages require a response. If you encounter an error that requires a response, review:

2.23 Responding to Errors

2.24 Out of Memory/Storage Space Errors

2.25 Setting the Directory Pathname



ATTENTION

If any error messages occur while using the Documentation Tool that are not listed in this subsection, contact the Honeywell Technical Assistance Center (TAC) at (800) 822- 7673.



ATTENTION

If multiple errors occur during query execution, it will all be appended to the "results file".

Messages

Documentation Tool error messages and their descriptions are as follows:

ILLEGAL AT BEG OF STRING

- Description: A repeat character (#) must have a character preceding it.

ARRAY PARAMETER USAGE INVALID

- Description: You cannot request an entire array by entering "array_name" in the "parameters to show port" for the Query command. Each element must be specified with its index. For example, PISRC is not legal, but PISRC(1), PISRC(2), or PISRC(3) is legal.

AT SIGN ILLEGAL AT END OF STRING

- Description: An escape character (@) is illegal at the end of a pattern.

CANNOT ADD DELETE FIELDS IN QUERY RESULTS OR PREBUILT QUERY DISPLAYS

- Description: You cannot use the <F10> function on query results or the Prebuilt Query display.

CANNOT DEFINE A FIELD ON A DISPLAY WITH SORT OR FILTER ON

- Description: Fields cannot be defined with the Sort or Filter commands turned on. Turn off Sort and Filter commands before attempting to define fields.

CANNOT SORT WHILE EDIT IS ON

- Description: You cannot sort a “delete prebuilt” display.

CURSOR IS NOT ON DISPLAY

- Description: The cursor is not intersecting a line of the display. When you press the <CTL> <F8> keys (Path), or select a prebuilt query, the cursor must intersect a line of data.

CURSOR WAS NOT ON A FIELD END COLUMN

- Description: The cursor must be at the end of the field column when deleting a field.

DATA TYPE CHANGED

- Description: This field contains more than one type of data; the data is not all numbers or date/time, so its data type has been changed to string.

DESCRIPTOR MUST BE USED TO SAVE QUERY PREBUILT

- Description: You must enter a descriptor in order to save a prebuilt query.

DUP FIELDS WITH DIFFERENT TYPES

- Description: You attempted to define a field with the same name as an existing field that has a different type defined for it.

DUPLICATE FIELD NAME EXISTS

- Description: You attempted to define another field with a name that already exists.

END OF INPUT

- Description: The end of data to be displayed has been reached.

EXCEEDS 240 CHARACTERS

- Description: Only 240 characters of a record can be displayed.

FAILURE

- Description: The operation has failed.

FIELD LENGTH ERR

- Description: The field name found in the “.DEFINE FIELD” command is greater than 16 characters in length, or the field type in the “.DEFINE FIELD” command is invalid.

FIELD DEF MISMATCH

- Description: You attempted to append an existing file with field definitions that do not match those already in the file. They must have the same name and type, and be in the same order.

FIELD NOT FOUND

- Description: The field you named and entered is not defined on the current display.

FILE ALREADY EXISTS

- Description: You must select the “overwrite” option if you want to output, but not append to an existing file.

FILE NAME OR EXTENSION EXPECTED

- Description: You must enter a file name or extension in the entry port.

FILE NOT FOUND

- Description:
 - The file specified in the Entity Names port does not exist or the file source (removable media) is removed during the query operation.
 - The sysqry.dc file is not present when you selected the recall operation (R680 and later).

FILE NOT LINKED TYPE

- Description: You cannot open a file unless its type is “L.” Use the List File Attributes (LS) command in the Command Processor to check the file type.

FILE PROTECT OR ACCESS PRIVILEGE VIOLATION

- Description: The file specified in the Entity Names port is already open.

FILTER IS OFF

- Description: You attempted to turn the Filter command off when it was already set to OFF.

HIGH LIMIT ERROR

- Description: The number you entered is greater than the maximum number allowed.

INDEX EXPECTED

- Description: The “(“ was not followed by the integer index.

INVALID CONDITIONAL EXPRESSION

- Description: The expression you entered is not in a valid format. Reference 2.21 Conditions for Query and Filter Commands.

INVALID DEVICE ID

- Description: The device ID you entered does not exist.

INVALID DEVICE WITH VOLUME ID

- Description: The device ID you entered does not exist.

INVALID FIELD ID

- Description: The field name specified using <CTL><F10> is too long or contains a delimiter (“or ,) or the field_ID specified in the .DEFINE_FIELD command is invalid.

INVALID FIELD ORDER

- Description: You attempted to edit the .DEFINE_FIELD definitions. Only one field of an undefined width (-32768) may be defined; it also must be last.

INVALID IDENTIFIER

- Description: Entity name port contains characters like ‘.’ or ‘>’ that are not preceded by NET or \$F.

INVALID PARAMETER ID

- Description: The parameter ID you entered does not exist.

INVALID PARAMETER INDEX

- Description: The index must be an integer within a range of valid indices.

INVALID PARAMETER OR FIELD NAME

- Description: The parameter or field name you entered does not exist.

INVALID PATHNAME

- Description: The pathname you entered is syntactically incorrect.

INVALID PATTERN SPECIFIER

- Description: The string you entered is not a legal pattern. Reference: Conditions for Query Journal

INVALID UNIT ID

- Description: While building a query, you entered a Unit ID that does not exist.

INVALID VALUE OR PATTERN

- Description: The value or pattern you entered is not valid. Reference: 2.21 Conditions for Query and Filter Commands or Conditions for Query Journal.

INVALID VERSION

- Description: Your version of the SYSQRY.DC file is incorrect.

INVALID VOLUME ID

- Description: You have entered an illegal character in the volume ID.

INVALID STARTING DATE-TIME ENTRY

- Description: You have entered an illegal character in the start date-time port.

INVALID ENDING DATE-TIME ENTRY

- Description: You have entered an illegal character in the end date-time port.

KEYLOCK MUST BE ENGINEER

- Description: The keylock must be ENGINEER in order to send output to a file.

MISSING DOT END COMMAND

- Description: The .DEFINE FIELD commands located at the start of a file with field definitions must be terminated by an .END command.

MISSING END BRACKET

- Description: The expression or condition is missing the end bracket.

MISSING RIGHT PAREN

- Description: The expression or condition is missing the right parenthesis.

MUST SELECT OVERWRITE TO SAVE

- Description: A prebuilt query with this descriptor name already exists. Press the [OVERWRITE] target or change the descriptor name.

NET>&DOC>SYSQUERY.DC FILE NOT FOUND

- Description: Displayed when a Doc Tool query is recalled and the &DOC directory is not available on HM (R680 and later).

NODE UNAVAILBLE

- Description: You attempted to do a query on a node that is presently down, a backup node, or a node that does not support a process entity.

NO DATA TO FIND

- Description: There is no data in the file that equals your Find command request.

NO DISPLAY TO CLOSE

- Description: There is no display presently open.

NO ERRORS TO DISPLAY

- Description: The <CTL> <F9> keys do not result in any errors to be displayed.

NO FILE EXISTS FOR THIS DISPLAY

- Description: There is no error display available, or the result of your request comes from memory, and not a temporary file.

NO MATCH FOUND

- Description: No data was found to match the specified criteria.

NO PREBUILT QUERIES TO DISPLAY

- Description:
 - You have not saved any prebuilt queries, so none can be called up.
 - Displayed when recall operation is exercised, when no saved query is available for recall (R680 and later).

NON-BLANK INPUT EXPECTED

- Description: Also applicable when the [FILTER] command from the main Doc Tool page is invoked, the state selected is ON but no condition is specified.

NON NUMERIC

- Description: An integer was required for this entry.

NOTHING TO FILTER

- Description: There is no data that applies to your Filter command request.

NOTHING TO OUTPUT

- Description: There is no data that can be output to either a file or printer.

NOTHING TO SORT

- Description: There is no data that applies to your Sort command request.

NUMERIC INPUT EXPECTED

- Description: You must enter a numeric value.

OPERATION ABORTED BY BREAK KEY

- Description: The operation was discontinued after using the<CTL> <BREAK> keys.

OPERATOR EXPECTED

- Description: A comparison or logical operator must be used in either the expression or condition.

OUT OF MEMORY

- Description: Available Documentation Tool memory has been exceeded.

OUT OF ROOM

- Description: Available Documentation Tool storage space has been exceeded.

OVERFLOW

- Description: The number you have specified is too large.

PATHNAME DOES NOT EXIST

- Description: The pathname you have entered does not exist.

PATTERN MUST BE SPECIFIED

- Description: You must enter a pattern within the condition or expression.
Reference: Conditions for Query Journal

PRESS ENTER TO EXECUTE COMMAND

- Description: You must press <ENTER> in order to execute a Documentation Tool command.

QUERY EXECUTION COMPLETE BUT NOT SAVED FOR RECALL

- Description: Displayed when the query execution is executed and not saved for recall due to the unavailability of SYSQUERY.DC file (R680 and later).

NOTE: This message is not displayed when the query encounters a run time error.

QUERY HAS NO DATA TO COLLECT

- Description: There were no entities returned from the query request.

REQUEST ERROR

- Description: Information you entered for your query request is incorrect.

REQUEST FAILED DUE TO HIWAY OR UCN UNAVAILABLE

- Description: Your request has failed because either the Data Hiway or UCN has shutdown or failed.

REQUEST FAILED DUE TO NODE UNAVAILABLE

- Description: Your request has failed because node has either shutdown or failed.

SINGLE IDENTIFIER EXPECTED

- Description: Multiple items are not allowed in this entry port.

SINGLE PATHNAME EXPECTED

- Description: Only one pathname for the desired file can be entered.

SINGLE VALUE EXPECTED

- Description: Multiple values are not allowed in this entry port.

SORT DATA CONTAINS BAD VALUES

- Description: Data that is in the sort field is in a format incompatible with the field's type. This can be either an invalid date, time, date/time combination, real, exponential, or integer format.

SYNTAX ERROR

- Description: Your request was syntactically incorrect.

SYS QUERY REQUIRES ENTITY TYPE

- Description: The Entity Names type cannot remain unselected while executing a query.

TOO MANY CHARACTERS

- Description: You have exceeded the number of characters for this entry port.

TOO MANY FIELDS

- Description: You have exceeded the number of fields that can be defined for this file.

TOO MANY RECORDS IN BLOCK FILE

- Description: The Documentation Tool cannot read this file.

USE OF WILDCARDS INVALID

- Description: The use of special character is not allowed with this command.

VALID PRINTER ID REQUIRED

- Description: The printer ID you entered to send output to is not valid.

VALUE OUT OF RANGE

- Description: The value that was entered was not within the legal range.

VOLUME SPECIFIED NOT FOUND

- Description: Directory specified does not exist.

VALID ONLY FOR JOURNAL QUERY

- Description: Displayed when the “Output to CSV” option is exercised for a non-journal query results file.

Index

D

- Data
 - Filtering, 32
 - Finding, 31
 - Sorting, 34
- Default Volume Pathnames, Modifying
 - Changing IDs to Removable Media, 17
 - Changing Individual Pathnames, 19
 - Preparations, 12
- Documentation Control directory
 - Creating, 6
- Documentation Tool
 - Conceptual diagram, 2
 - Error Messages
 - Messages, 84
 - Help Function, 77
 - Directory Pathnames, 83
 - Typical users, 3, 4
 - Using, 11

E

- Errors
 - Out of Memory/Storage Space, 80

F

- Fields
 - Changing, 41
 - Defining, 41
 - Deleting, 41
- Files
 - Closing, 30
 - Outputting, 38
 - Save options, 38
- Filter Command
 - Filter state, 32
 - Procedure, 32
 - State and Condition entry box, 33
- Filter Command Definition, 32
- Finding Point Interconnections, 51

K

- Keylock access, 17
 - Outputting to a file, 38

P

- Parameters occurring on every point, 63
- point interconnection function, 51

Q

- Queries with Selection List, 63
- Query
 - Building, 43
 - Commands, 23
 - Patterns for
 - Find, Patterns for
 - Pattern for, 75
 - Errors, 24
 - Examples, 22
 - Executing, 43
 - Conditions for
 - Conditions for, 70
 - Operations, 22
 - Prebuilt Query, Deleting, 69
 - Prebuilt Query, Executing, 67
 - Prebuilt Query, Modifying, 67
 - Recalling, 43
 - Results, 23
 - Rules for Entering Setup Data, 61
 - Saving, 43
 - Setup example, 23
- Query Journal Events, 48, 51

R

- Retrieval Filter for Journal Queries, 49

S

- Sorting Journal Queries, 36

T

- Temporary File directory
 - Creating, 6

Process Solutions

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